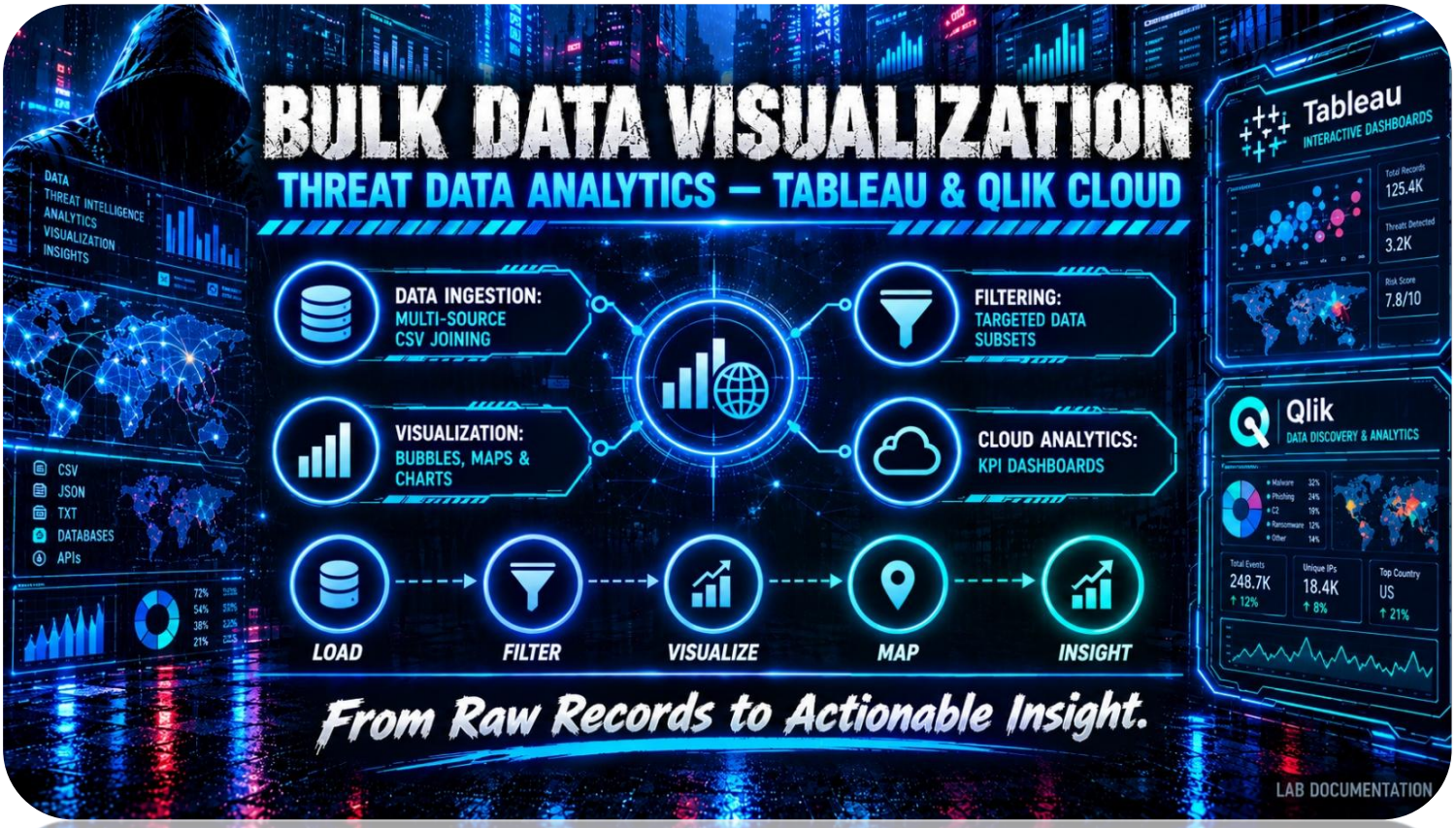


Abdul Wahid

Lab Report

Healthcare Breach Data Analysis – Tableau & Qlik Cloud

Executive Summary

This lab demonstrates bulk data management and visualization as part of the Certified Threat Intelligence Analyst (CTIA) curriculum, Module 04 (Data Collection and Processing). Two industry-standard analytics platforms were used to analyze the HHS Cyber Security Breaches dataset, which records healthcare data breaches reported to the US Department of Health and Human Services.

Tableau Desktop (Part 1): Tableau Desktop 2018.1 was installed and activated under a 14-day trial. Two CSV datasets (HHSCyberSecurityBreaches.csv and breaches.csv) were loaded and combined with an Inner Join on the F1 field. A data source filter restricted records to 1,500–2,000 individuals affected, returning 90 rows. After the filter was removed, geographic, Packed Bubbles, and Stacked Bar visualizations were built to compare breach impact by state. Total individuals affected across the dataset: 37,773,276.

Qlik Cloud Analytics (Part 2): A Qlik Cloud Analytics trial was activated, and the Healthcare_Breach_Analysis application was created in the Personal space. The HHSCyberSecurityBreaches.csv dataset was uploaded (approximately 1,150 records), and a sheet titled "Total Individuals Affected" was configured with a KPI chart (record count) and a Map chart.

Key Findings: Tennessee recorded the highest total of individuals affected, followed by Indiana/Illinois, New York, and Texas. Visual analysis surfaced geographic concentration patterns that are difficult to detect in raw tabular data.

Outcome: The lab shows how threat intelligence analysts can ingest multi-source bulk data, join and filter it, and turn it into visual insight using both a local desktop tool and a cloud-based platform.

Contents

Bulk Data Management & Visualization Using Tableau 4

Section 1.1 – Accessing the Tableau Website 4

Section 1.2 – Installing Tableau Desktop 5

Section 1.3 – Activating Tableau (Trial Registration) 7

Section 1.4 – Tableau Main Interface and Data Connection 10

Section 1.5 – Loading the HHSCyberSecurityBreaches Dataset 11

Section 1.6 – Adding the Second Dataset (breaches.csv) 12

Section 1.7 – Configuring the Join Between Datasets 14

Section 1.8 – Applying Data Source Filters 14

Section 1.Y – Viewing the Filtered Dataset 16

Section 1.10 – Removing Filter and Navigating to Worksheet 17

Section 1.11 – Adding State Dimension to Worksheet 18

Section 1.12 – Individuals Affected Visualization (Packed Bubbles) 20

Section 1.13 – Stacked Bar Chart Visualization 22

Part 2: Data Visualization Using Qlik Cloud Analytics 23

Section 2.1 – Accessing the Qlik Website and Trial Options 23

Section 2.2 – Creating a Qlik Account (Password Setup) 24

Section 2.3 – Qlik Cloud Analytics Home Dashboard 25

Section 2.4 – Creating a New Qlik Application 26

Section 2.5 – Loading Data into Qlik 26

Section 2.6 – Choosing Analytics Options in Qlik 28

Section 2.7 – Qlik Sheet Editor 29

Section 2.8 – Adding KPI Chart and Measures 30

Section 2.Y – Configuring Sheet Title and Properties 32

Section 2.10 – Qlik Cloud Analytics Capabilities 32

Lab Conclusion and Summary 34

Key Accomplishments 34

Learning Outcomes 34

Bulk Data Management & Visualization Using Tableau

Tableau is one of the leading bulk data management and visualization tools used to manage and visualize collected data for building threat intelligence. In this part, we install Tableau Desktop, load the HHS Cyber Security Breaches datasets, apply filters, and create visualizations.

Section 1.1 – Accessing the Tableau Website

The first step is to visit the official Tableau website to download and access the product. The website provides options for Tableau Desktop (local install), Tableau Prep Builder, and Tableau Server.

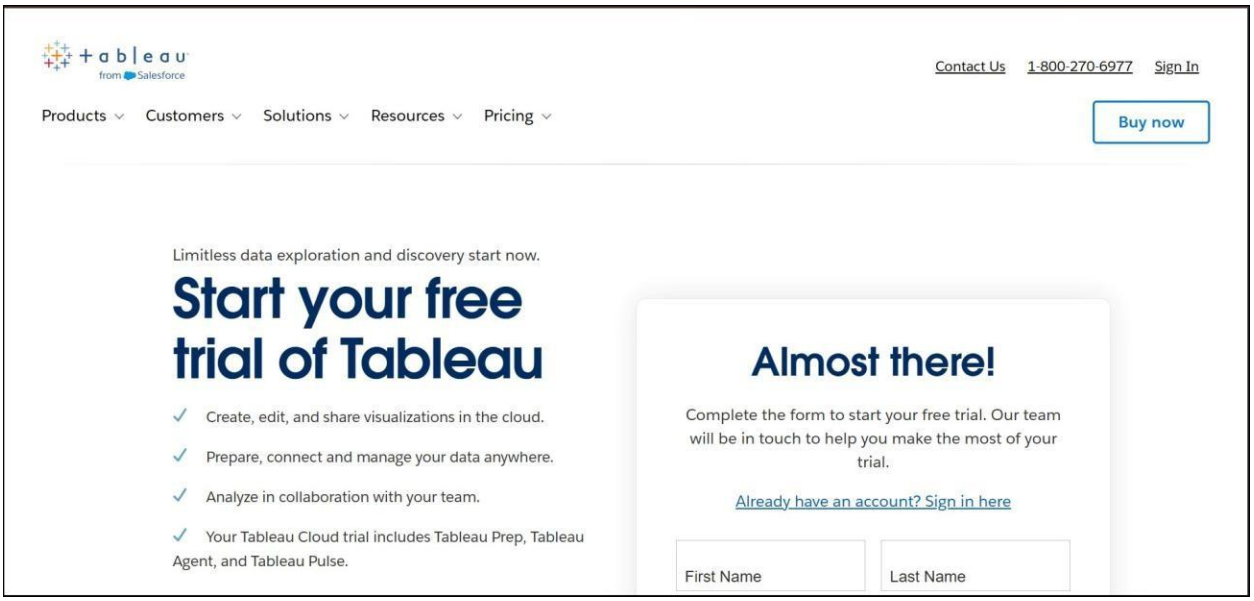


Figure 1: Tableau official website – "Start your free trial of Tableau" page showing the registration form.

The Tableau website (tableau.com) displays the free trial signup page. It highlights key features: create/edit/share visualizations in the cloud, prepare and connect data, and analyze in collaboration with your team. The free trial includes Tableau Prep, Tableau Agent, and Tableau Pulse.



Figure 2: Tableau website footer showing product categories including Tableau Desktop, Tableau Prep Builder, and Tableau Server, along with community and support links.

The footer of the Tableau website lists multiple product options. For this lab, Tableau Desktop was selected as it enables building powerful calculations, views, and dashboards locally on your computer.

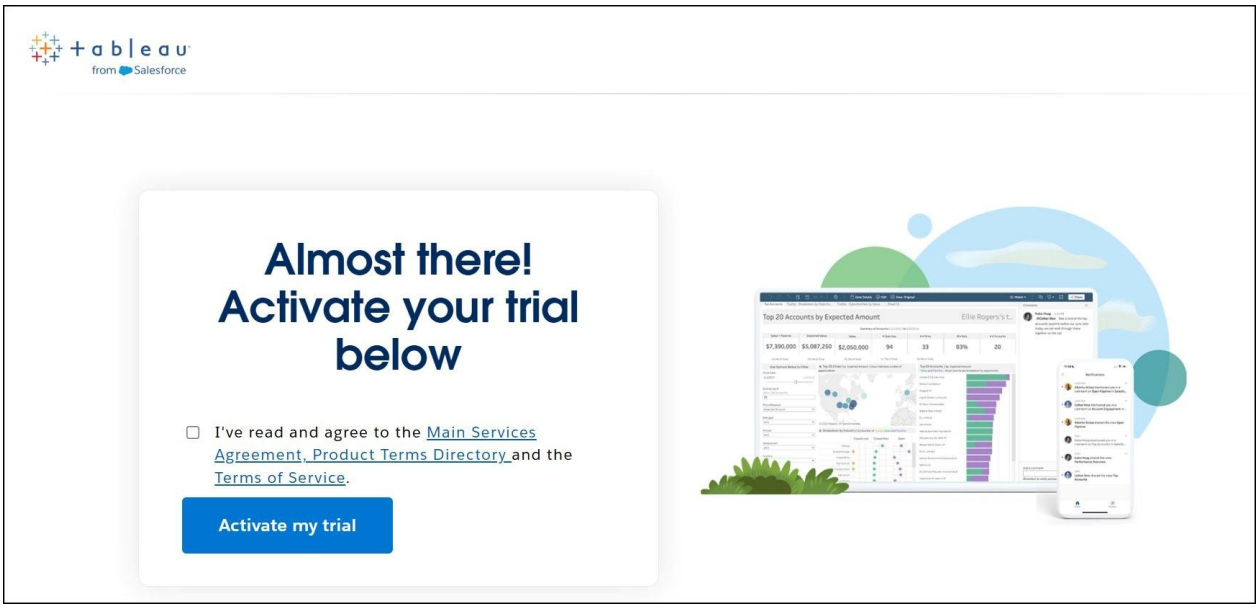


Figure 3: Tableau product page showing three options – Tableau Desktop (Download for free), Tableau Prep Builder (Download a free trial), and Tableau Server (Contact Sales).

From the product page, Tableau Desktop was chosen. It can be downloaded for free as a trial and allows local data analysis and dashboard creation. This is the tool used throughout Part 1 of the lab.

Section 1.2 – Installing Tableau Desktop

After downloading Tableau Desktop 2018.1, the installation wizard guides the user through the setup process.

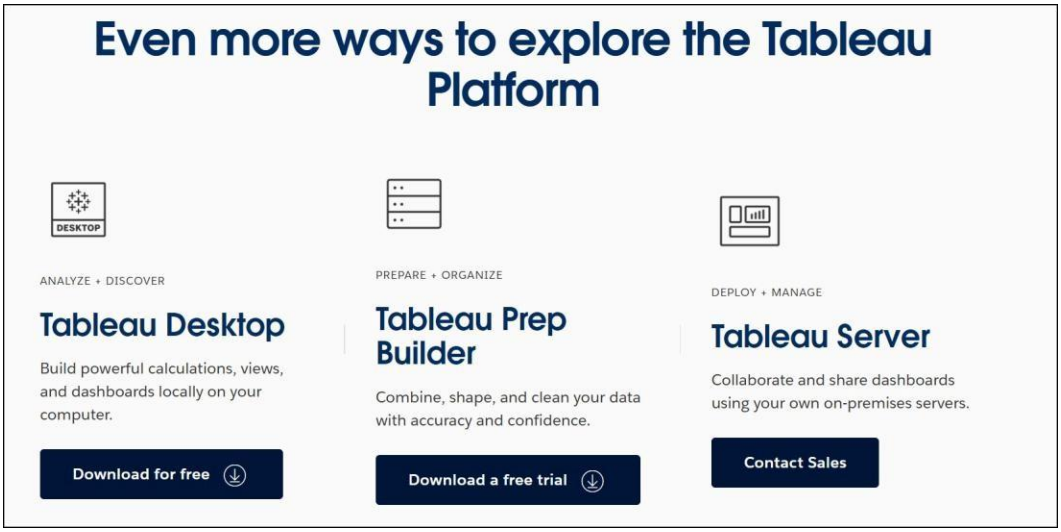


Figure 4: Tableau 2018.1 Setup window – The license agreement must be accepted before installation can proceed.

The Tableau Setup window displays the Tableau 2018.1 license terms. The user must check the "I have read and accept the terms of the license agreement" checkbox before the Install button becomes active.

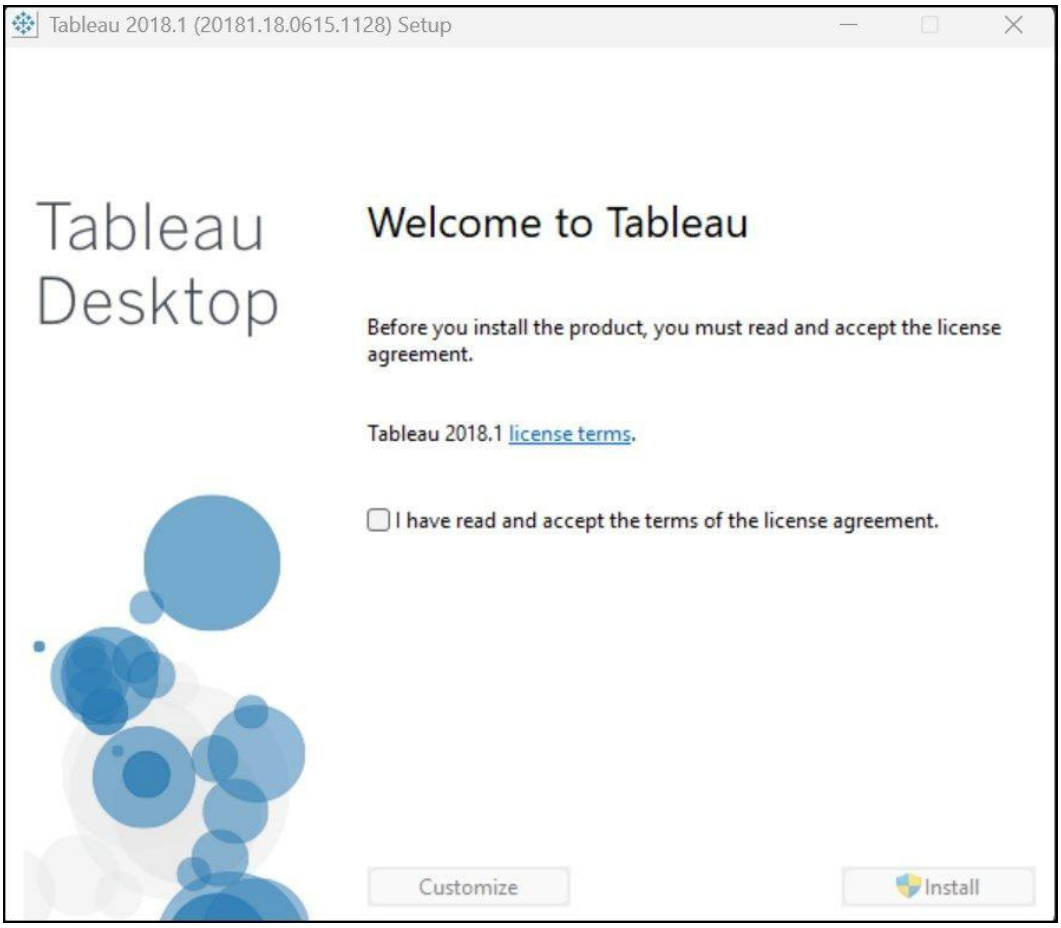


Figure 5: Tableau 2018.1 Setup – License agreement checkbox unchecked (Install button is greyed out).

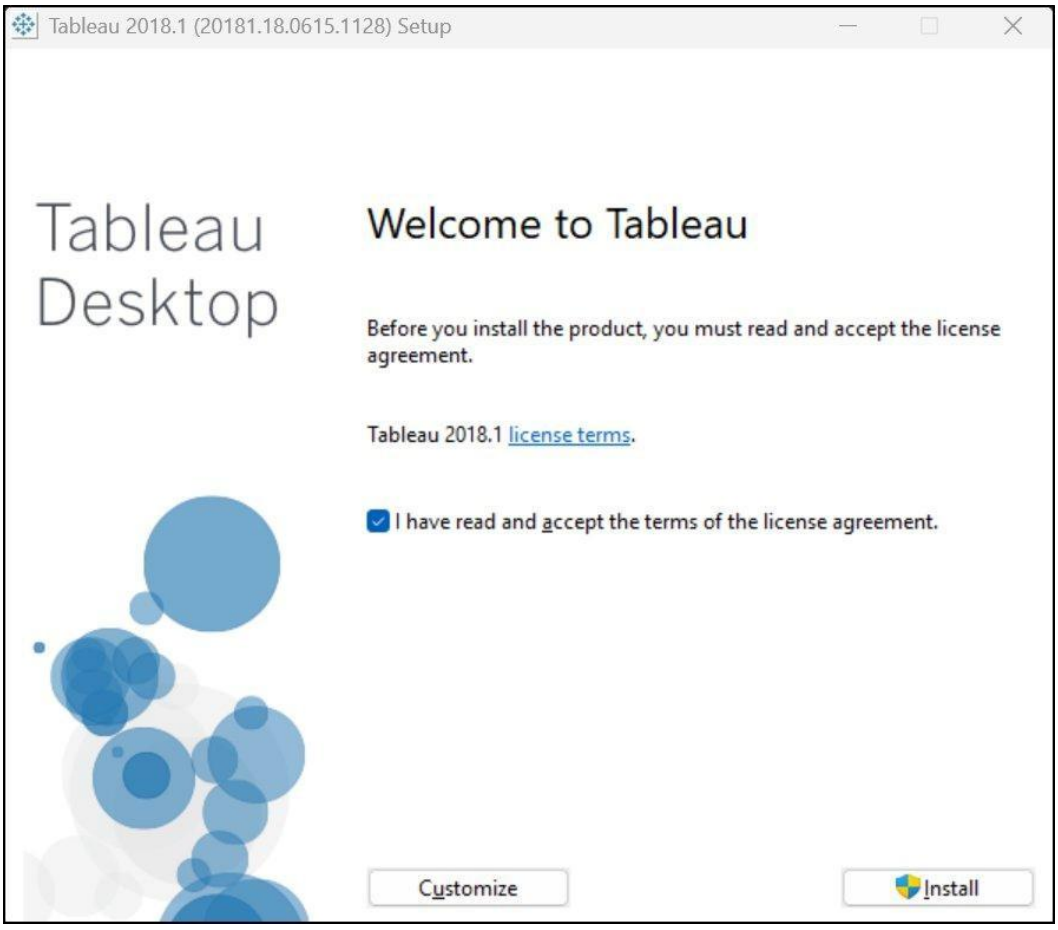


Figure 6: Tableau 2018.1 Setup – License agreement checkbox checked (Install button is now active and ready to proceed).

Once the license agreement checkbox is checked, the Install button becomes active. Clicking it begins the installation process. The wizard proceeds with default options, and after installation, a restart of the machine is required.

Section 1.3 – Activating Tableau (Trial Registration)

After installation and restart, Tableau launches and displays the Activate Tableau window. The user selects "Start trial now" to begin the 14-day free trial without restriction.

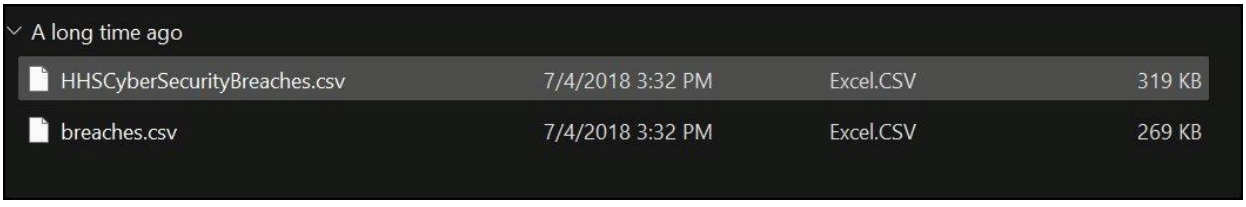


Figure 7: Two CSV data files visible in the Bulk Data folder – HHSCyberSecurityBreaches.csv (319 KB) and breaches.csv (269 KB). These are the datasets used in this lab.

The Bulk Data folder contains the two datasets used in this lab. Both files are in Excel CSV format and were provided as part of the CTIA course materials.

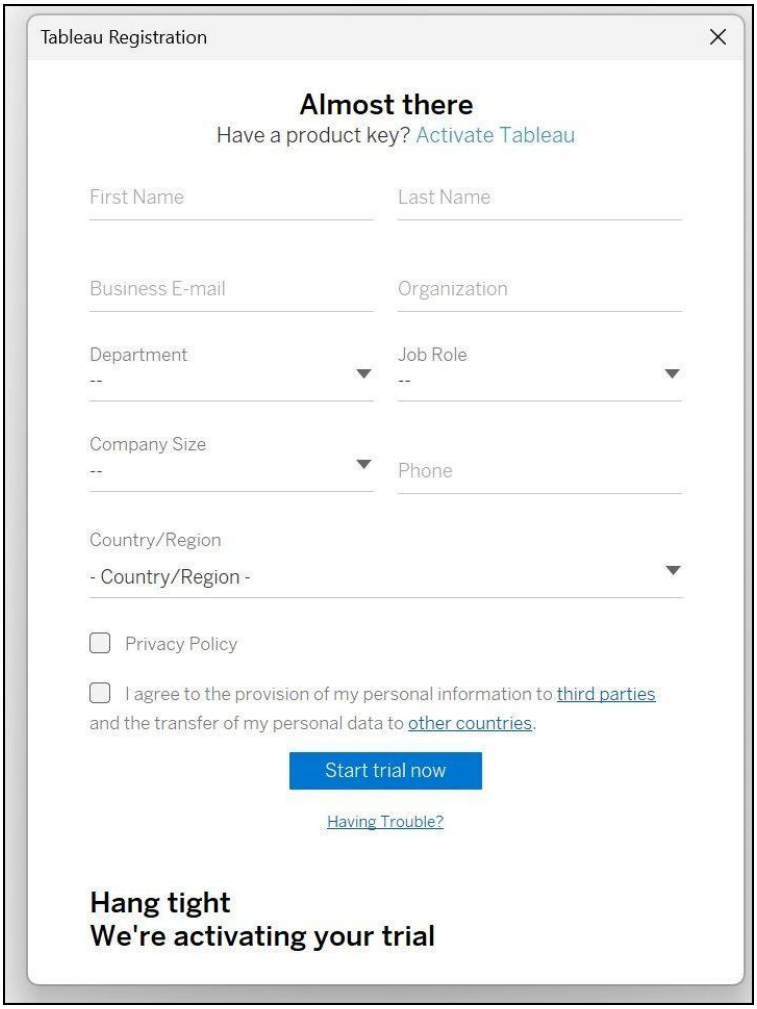


Figure 8: Tableau Activation window showing "Start trial now" option (14 days without restriction) and "Activate" option (enter product key).

The Activate Tableau window presents two choices: (1) Start trial now – Use Tableau for 14 days without restriction, or (2) Activate – Enter a product key to activate Tableau. For this lab, "Start trial now" is selected.

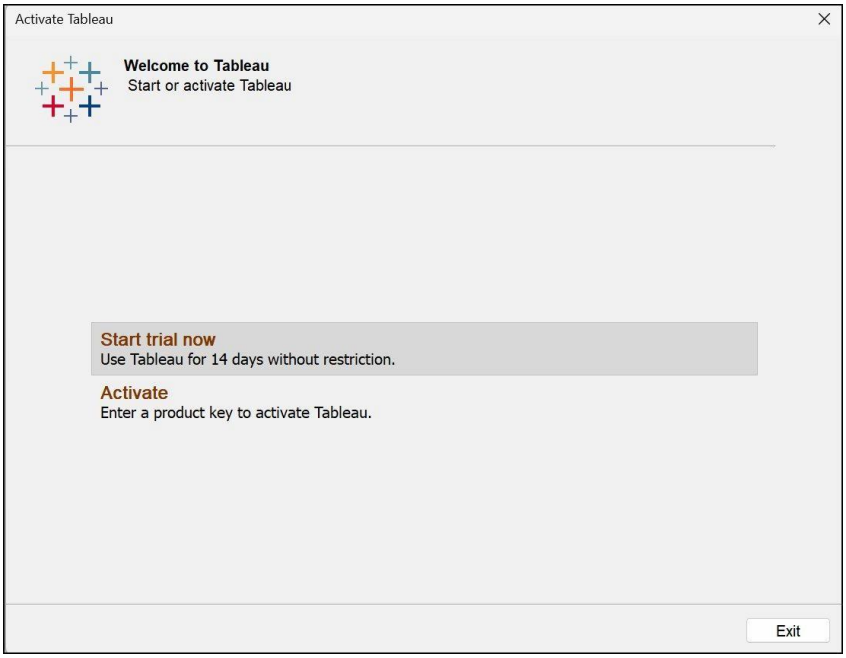


Figure 9: Tableau Registration form – User fills in First Name, Last Name, Organization, Email, Phone, Job Title, City, Postal Code, Country/Region, Department, and Industry.

The Registration page requires complete user information. The details filled in are: Name: Abdul Wahid, Organization: Global Tech Ltd, Email: 243699@students.au.edu.pk, Phone: +92346420393, Job Title: Analyst, City: Multan, Country: Pakistan, Department: Other, Industry: Other.

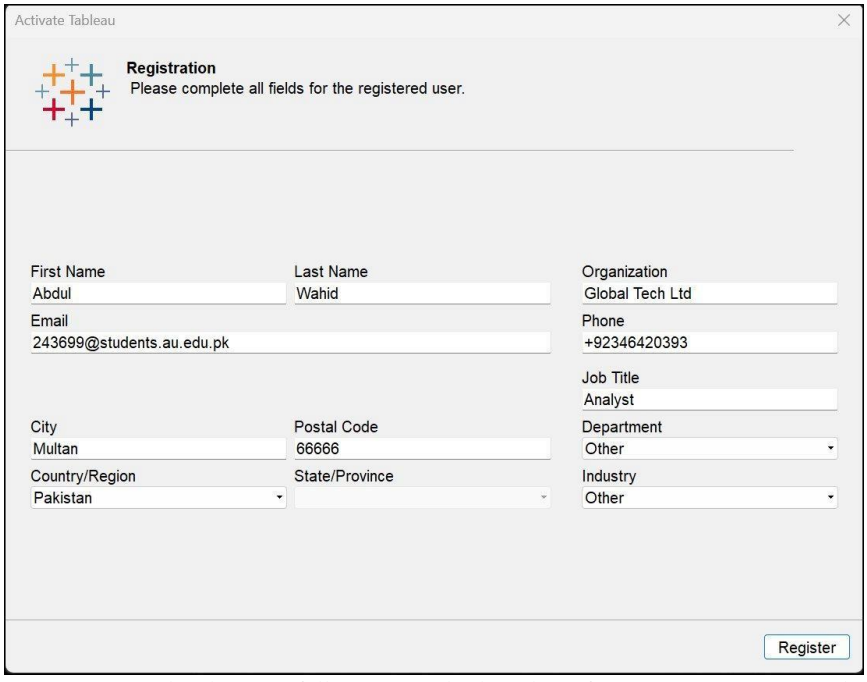


Figure 10: Tableau Registration completed successfully – Green checkmark confirms "Registration completed." with a Continue button to proceed.

After submitting the registration form, a green checkmark confirmation appears: "Registration completed." The user clicks Continue to proceed to the main Tableau interface.

Section 1.4 – Tableau Main Interface and Data Connection

After successful registration, the Tableau Book1 window opens. The Connect panel on the left shows options to connect to files (Excel, Text, JSON, etc.) and to servers.



Figure 11: Tableau Book1 – Main interface with Connect panel on the left showing "To a File" and "To a Server" options, and Sample Workbooks in the center. The 14-day trial welcome message is visible.

The Tableau main window shows the Connect panel with multiple data source options. The "To a File" section includes Microsoft Excel, Text file, JSON file, and more. The center panel shows an Open area and Sample Workbooks. A welcome pop-up confirms the 14-day trial has started.

Section 1.5 – Loading the HHSCyberSecurityBreaches Dataset

The first dataset, HHSCyberSecurityBreaches.csv, is loaded by navigating to the Bulk Data folder and opening the file. This is done via the "Text file" option in the Connect panel.

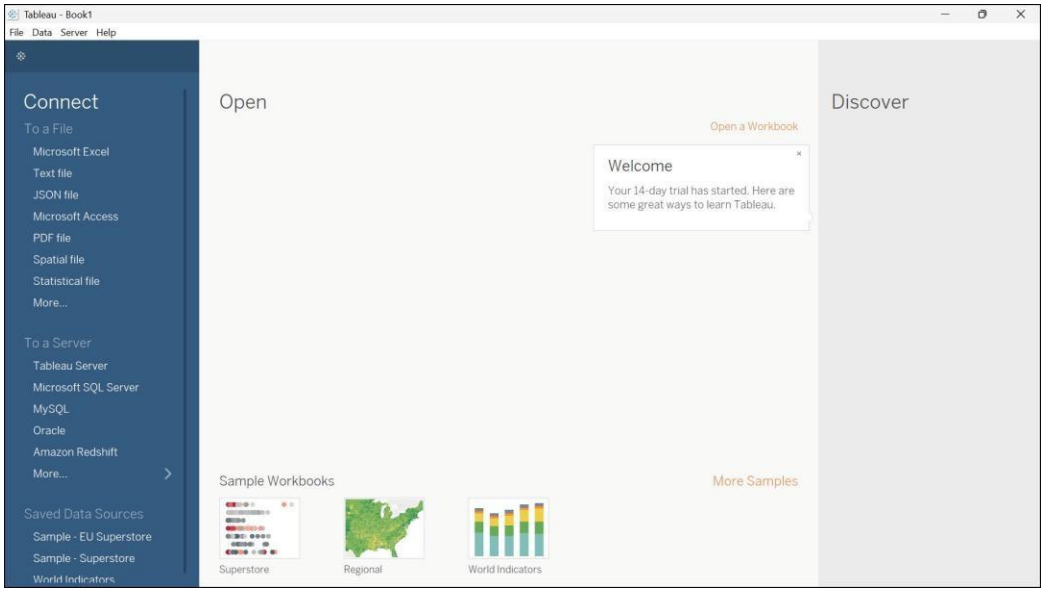


Figure 12: File Open dialog showing the Bulk Data folder with HHSCyberSecurityBreaches.csv selected. The file name is confirmed in the filename field.

Using the "Text file" option, the Open dialog navigates to the Downloads > Bulk Data folder. The file HHSCyberSecurityBreaches.csv is selected and opened.

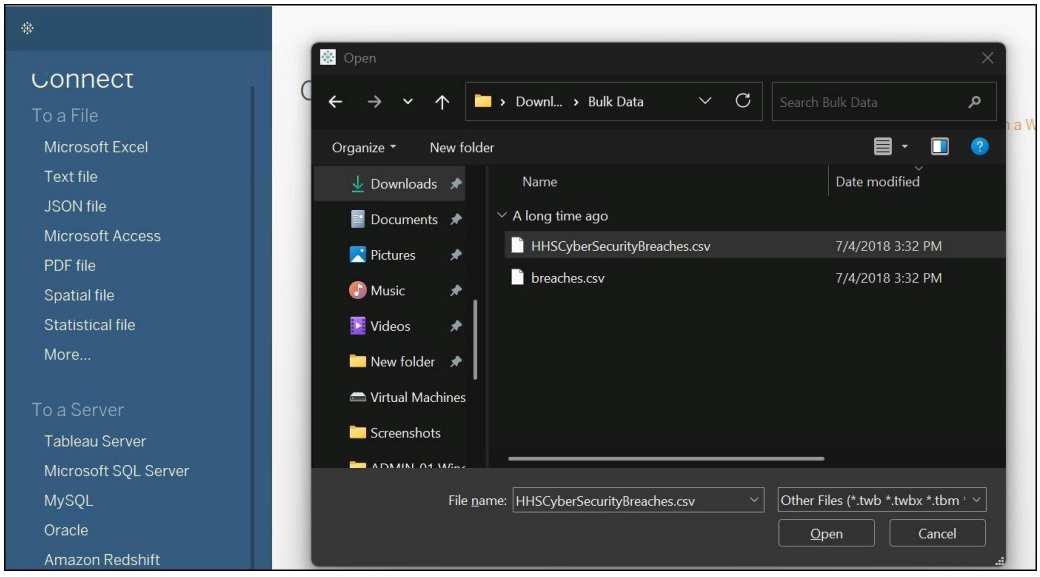


Figure 13: Tableau Data Source view showing the HHSCyberSecurityBreaches dataset loaded. The table preview shows columns: F1, Name of Covered Entity, State, Covered Entity Type, Individuals Affected, Breach Submission Date, Type of Breach, Location of Breached Information, Business Associate Involved, and Web Description.

After loading, the Data Source view displays the dataset in a spreadsheet preview. Key columns include: Name of Covered Entity, State, Covered Entity Type, Individuals Affected, Breach Submission Date, Type of Breach, and Location of Breached Information. The first few records show healthcare providers from TX, MO, AK, DC, CA states, with breach types including Theft and Loss.

Section 1.6 – Adding the Second Dataset (breaches.csv)

To perform a multi-dataset analysis, a second dataset (breaches.csv) is added to the project by clicking the "Add" button in the Connections panel and selecting another text file.

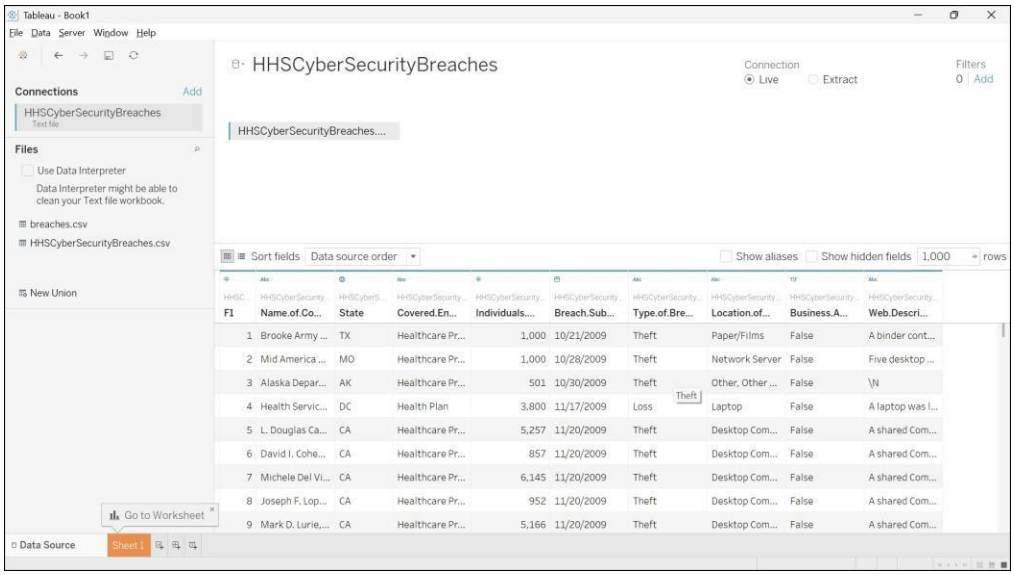


Figure 14: The "Add a Connection" dialog appears with options to connect "To a File" (Microsoft Excel, Text file, JSON file, Microsoft Access, PDF file, Spatial file, Statistical file, More...) and "To a Server" options.

Clicking "Add" in the Connections panel opens the "Add a Connection" menu. Under the "To a File" section, "More..." is clicked to access additional file format options. The list on the right shows various server-based connectors like Actian Matrix, Amazon Athena, Amazon Redshift, and others.

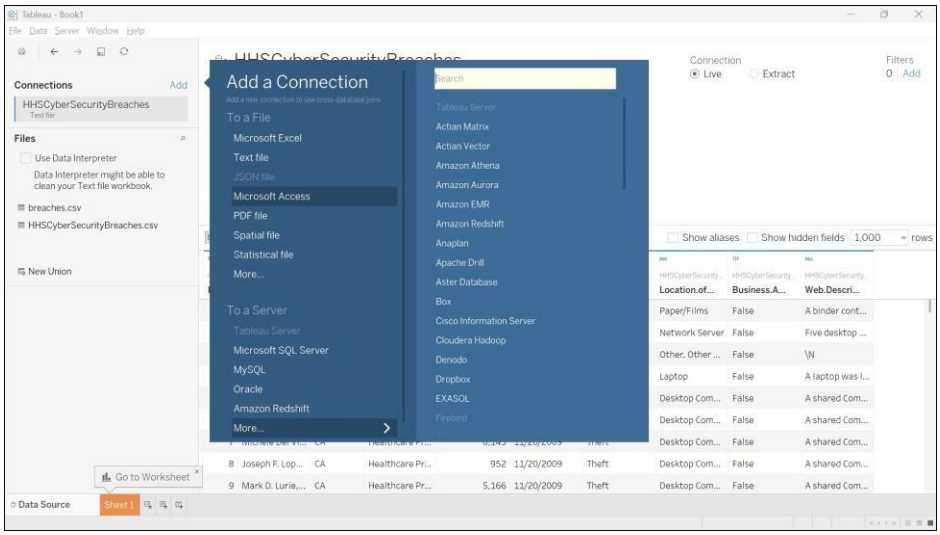


Figure 15: The Open file dialog showing the Bulk Data folder with breaches.csv highlighted and selected. Both CSV files are visible: HHSCyberSecurityBreaches.csv and breaches.csv.

In the Open dialog, the Bulk Data folder is navigated to. The breaches.csv file is selected (shown highlighted in the file list). The filename is confirmed in the File name field and the Open button is clicked.

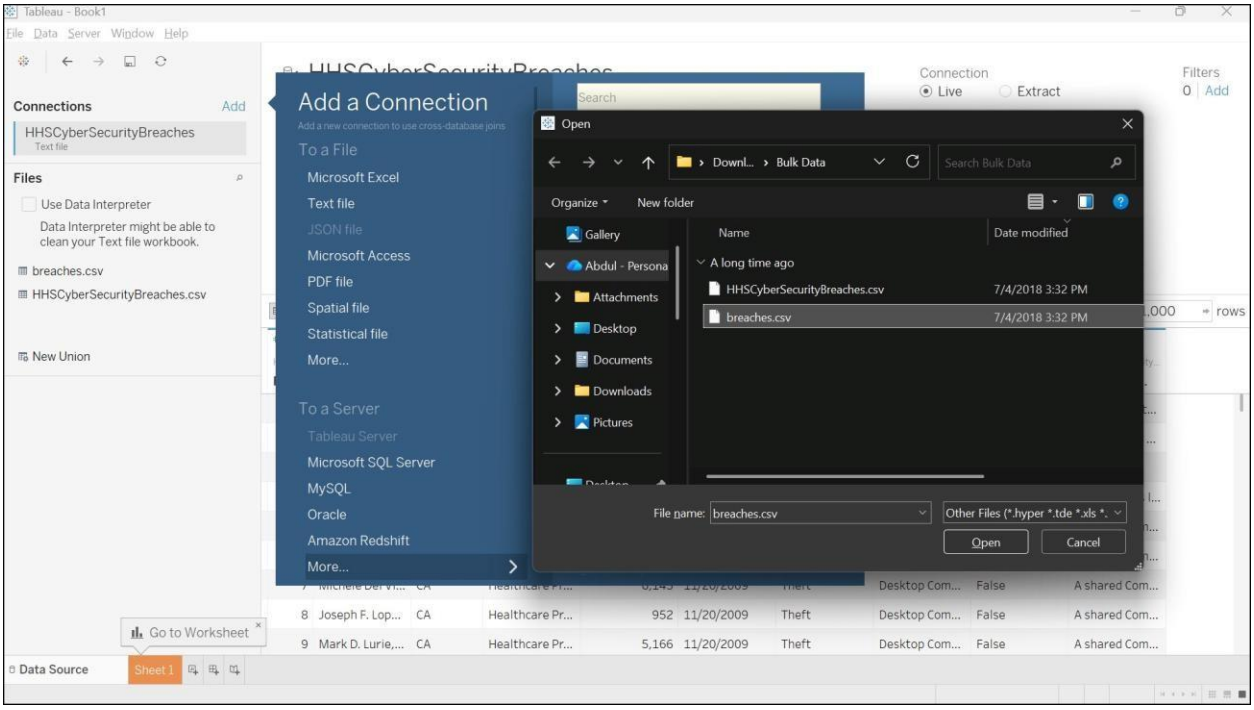


Figure 16: Tableau Data Source view after adding the second dataset – The title bar now shows "HHSCyberSecurityBreaches.csv+ (Multiple Connect...)" and both connections appear in the Connections panel. The visual connector shows the two files linked with an Inner Join symbol.

After adding breaches.csv, the data source name updates to show "(Multiple Connect...)" indicating multiple connections. Both "HHSCyberSecurityBreaches" and "breaches" appear in the Connections panel on the left. The central canvas shows the two files joined together with an Inner Join by default.

Section 1.7 – Configuring the Join Between Datasets

Tableau automatically creates an Inner Join between the two datasets on the F1 field. The Join options (Inner, Left, Right, Full Outer) can be configured by clicking the join symbol.



Figure 17: The Join configuration popup showing all four join types: Inner (selected), Left, Right, and Full Outer. The join clause shows F1 = F1 (breaches.csv), meaning both datasets are joined on the F1 (row number) field.

The Join dialog shows four join types available. The Inner join is selected (highlighted in blue), which means only matching records from both datasets will be included. The join clause is configured as: Data Source (F1) = breaches.csv (F1), meaning the datasets are joined on their row index fields.

Section 1.8 – Applying Data Source Filters

Data source filters allow the analyst to narrow down the dataset based on specific criteria. The filter is applied by clicking the "Add" link next to Filters in the top-right corner of the Data Source view.

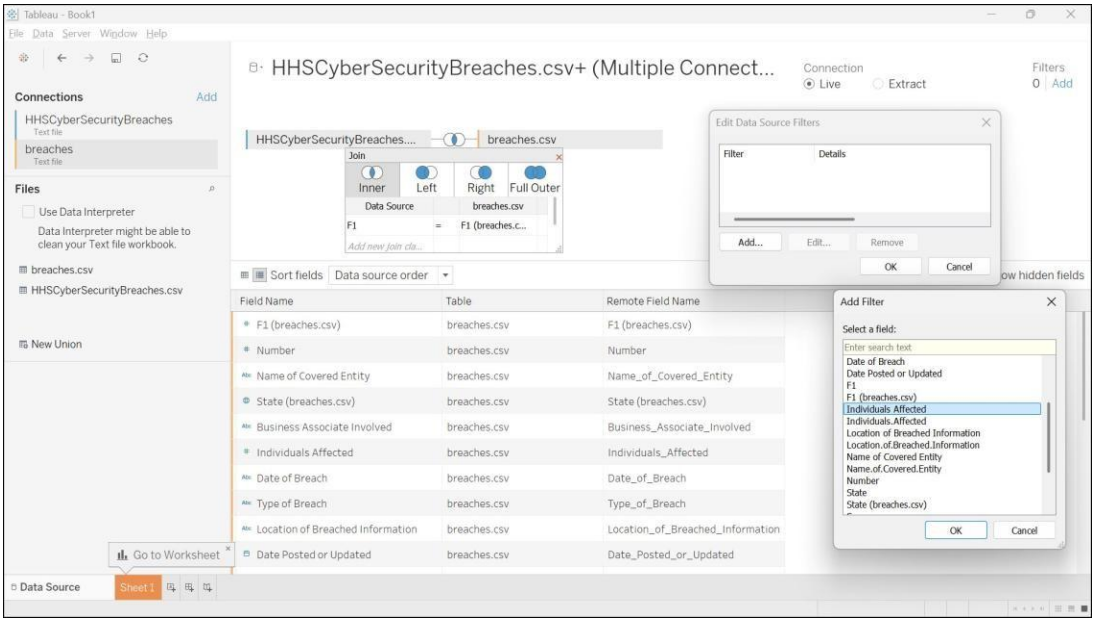


Figure 18: The Edit Data Source Filters dialog and Add Filter selection list. The "Individuals Affected" field is highlighted in the Add Filter list, ready to be selected as the filter field.

The Edit Data Source Filters dialog appears after clicking "Add" under Filters. The Add Filter list shows all available fields including: Date of Breach, Date Posted or Updated, F1, Individuals Affected, Location of Breached Information, Name of Covered Entity, Number, State, and others. "Individuals Affected" is selected as the filter field.

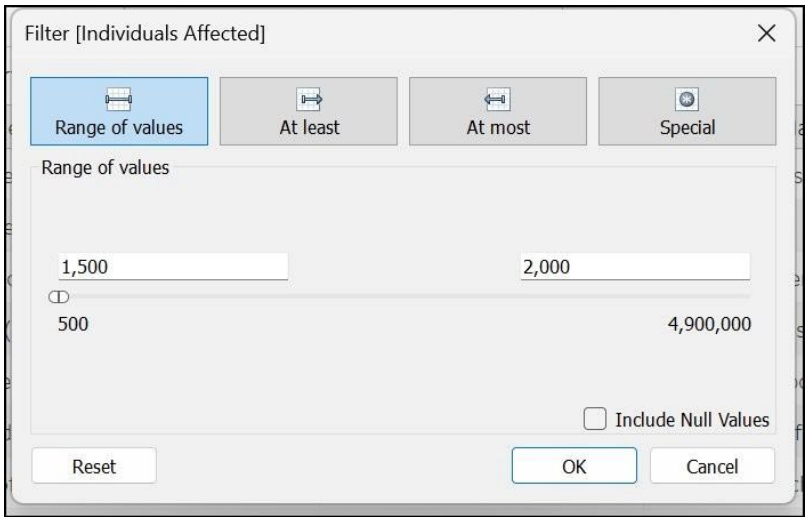


Figure 19: Filter [Individuals Affected] dialog – "Range of values" tab selected, with minimum value set to 1,500 and maximum value set to 2,000. The full data range spans from 500 to 4,900,000.

In the Filter [Individuals Affected] window, the "Range of values" tab is selected. The range is set from 1,500 (minimum) to 2,000 (maximum). This filters the dataset to only show breach records where the number of individuals affected falls between 1,500 and 2,000. The OK button is clicked to apply.

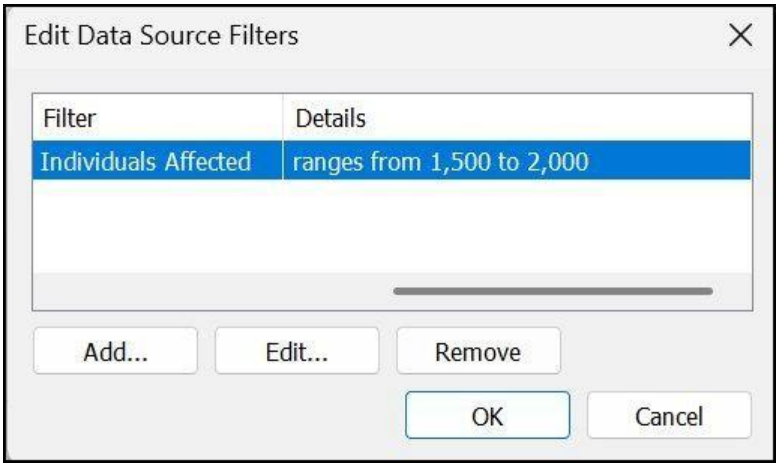


Figure 20: Edit Data Source Filters confirmation window showing the applied filter – "Individuals Affected" with detail "ranges from 1,500 to 2,000".

The Edit Data Source Filters window confirms the filter has been applied: "Individuals Affected – ranges from 1,500 to 2,000". Clicking OK applies this filter to the entire data source, and only records meeting this criterion will appear in the worksheet.

Section 1.Y – Viewing the Filtered Dataset

After applying the filter, the data source view refreshes to show only records where Individuals Affected is between 1,500 and 2,000.

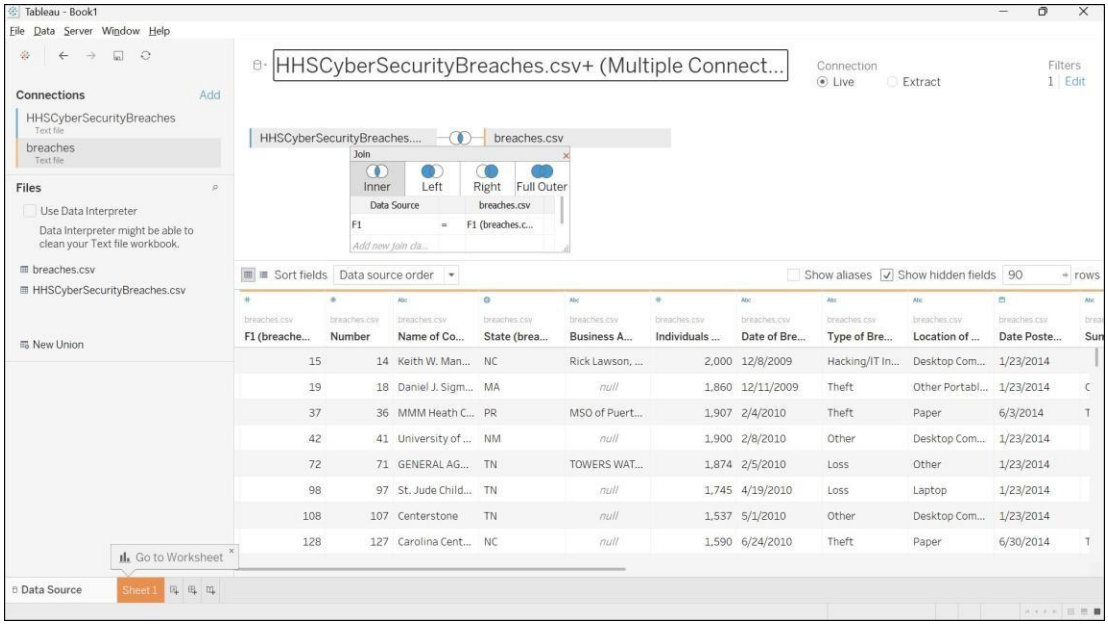


Figure 21: Filtered data view showing records with Individuals Affected between 1,500 and 2,000. The filter count shows 1 filter applied (Filters: 1). Visible records include healthcare entities from NC, MA, PR, NM, TN states.

The filtered dataset shows 90 rows (as indicated in the top-right). Records visible include: Keith W. Mann (NC, 2,000 individuals), Daniel J. Sigman MD PC (MA, 1,860), MMM Heath Care Inc (PR, 1,907), University of New Mexico (NM, 1,900), GENERAL AGENCIES (TN, 1,874), St. Jude Children's Research Hospital (TN, 1,745), and Centerstone (TN, 1,537). The Join popup remains visible showing the Inner join configuration.

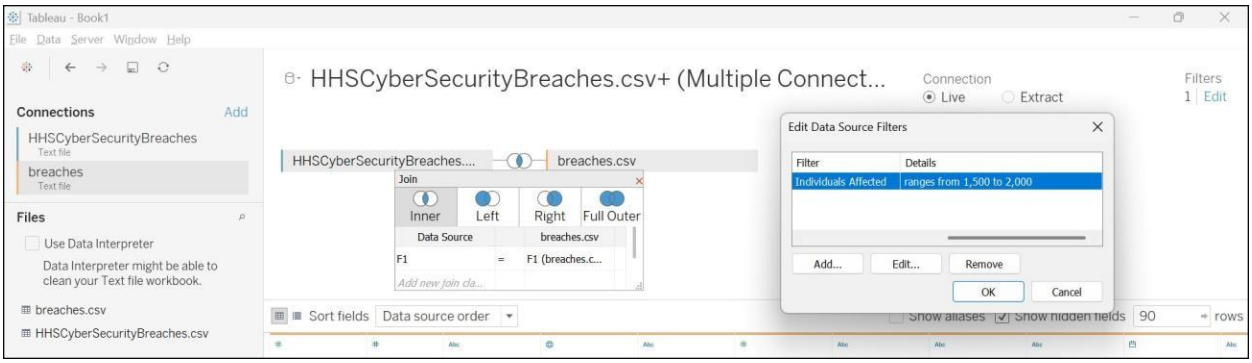


Figure 22: Edit Data Source Filters window showing the confirmed filter – "Individuals Affected ranges from 1,500 to 2,000" highlighted in the filter list.

The Edit Data Source Filters dialog confirms the active filter. The Individuals Affected filter with range 1,500 to 2,000 is shown highlighted. The Add, Edit, and Remove buttons allow modification of filters. Clicking OK applies the filter.

Section 1.10 – Removing Filter and Navigating to Worksheet

After reviewing filtered data, the filter is removed to restore the full dataset for visualization. The filter icon (Edit) is clicked, the filter is selected, and Remove is clicked. Then the worksheet is opened.



Figure 23: Tableau interface after filter removal – The title shows "HHSCyberSecurityBreaches.csv+ (Multiple Connect...)" with the two datasets joined. Filters count returns to 0 after removal.

After removing the Individuals Affected filter, the data source reverts to showing the full unfiltered dataset. The visual join diagram shows the two files (HHSCyberSecurityBreaches and breaches.csv) connected with an Inner Join. Both connections remain active in the Connections panel.

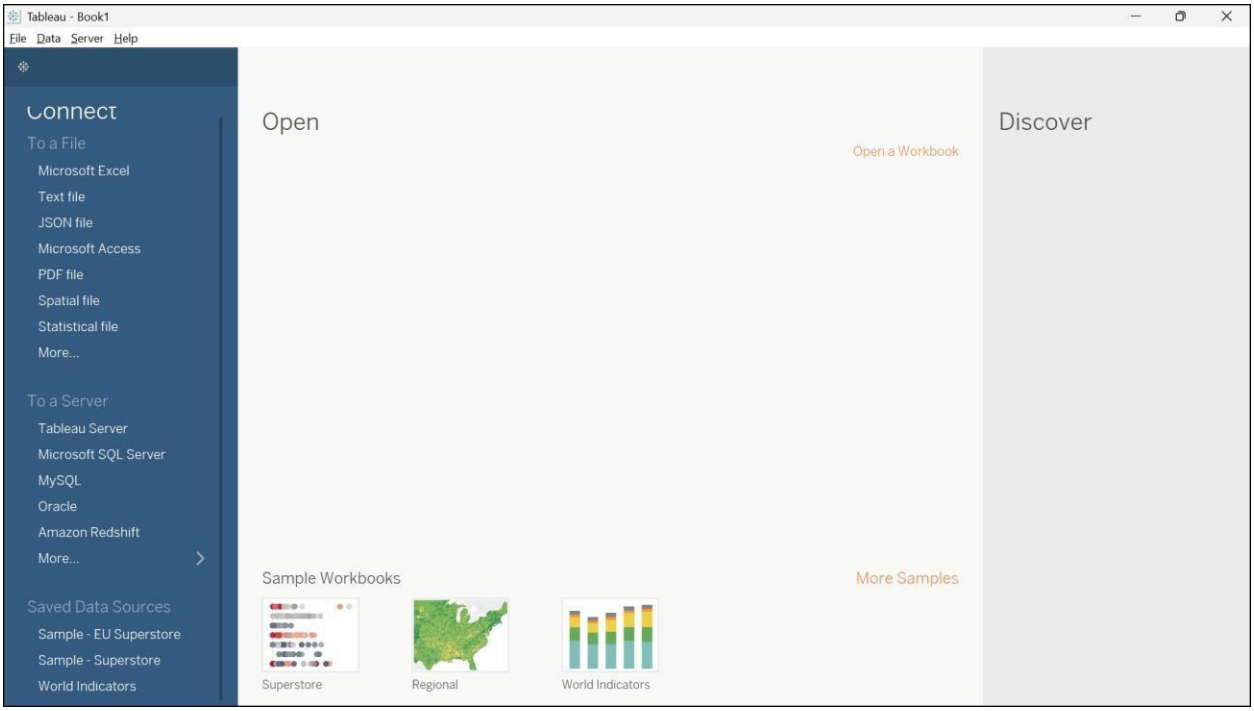


Figure 24: Tableau Book1 main Connect panel – The interface shows the full Connect panel with "To a File", "To a Server", and "Saved Data Sources" sections. World Indicators is visible in Saved Data Sources.

The main Tableau Connect panel is visible. Under "Saved Data Sources", World Indicators is available as a pre-built sample. This will be used to provide geographic context for visualizations in the following steps.

Section 1.11 – Adding State Dimension to Worksheet

In the worksheet, the State dimension from the HHSCyberSecurityBreaches data is added to the sheet to begin geographic visualization.

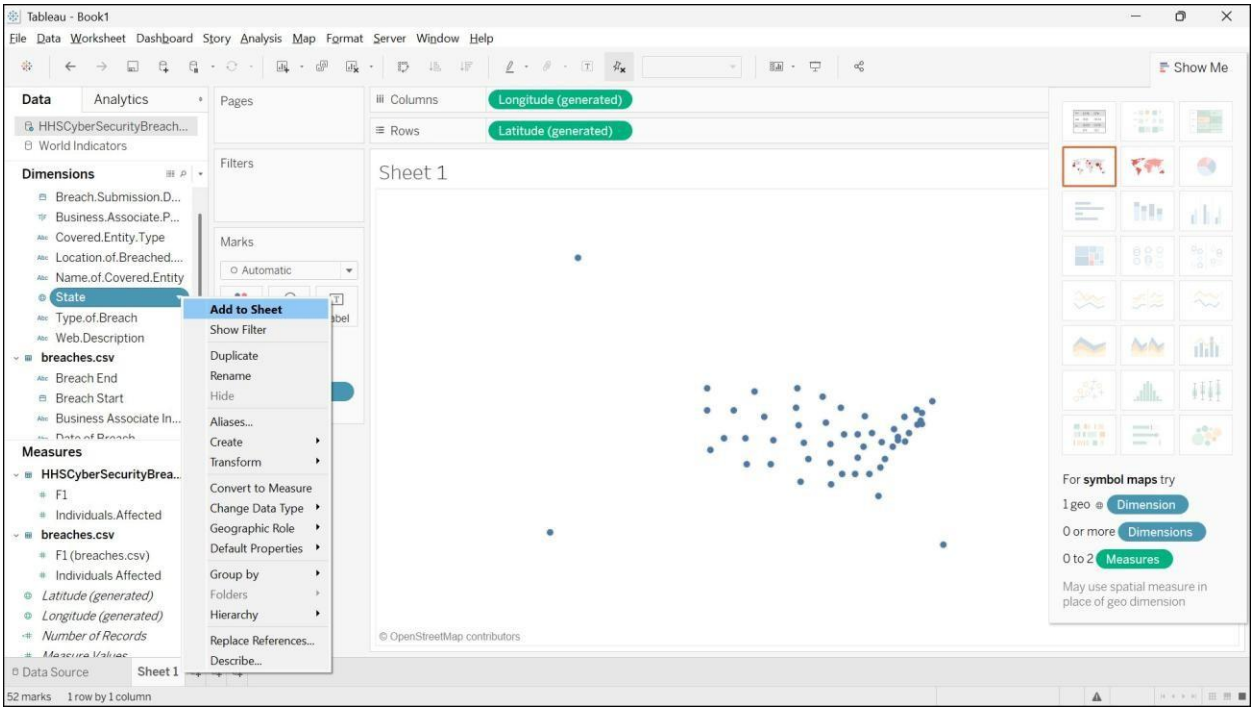


Figure 25: Sheet 1 worksheet view showing the State dimension right-click menu. The "Add to Sheet" option is highlighted. The Dimensions panel shows fields from both HHSCyberSecurityBreaches and breaches.csv.

In the Data panel on the left, the State field under HHSCyberSecurityBreaches dimensions is right-clicked. The context menu shows options including "Add to Sheet", Show Filter, Duplicate, Rename, Hide, Aliases, Create, Transform, Convert to Measure, Change Data Type, Geographic Role, Default Properties, Group by, Folders, Hierarchy, Replace References, and Describe. "Add to Sheet" is selected.

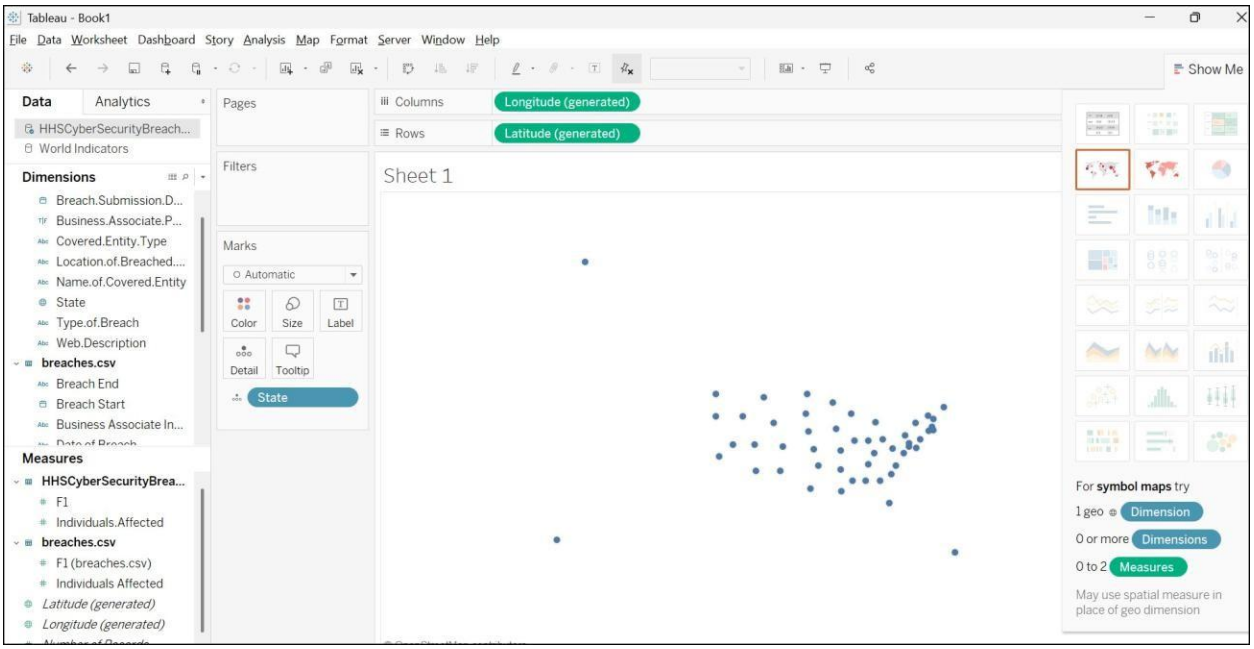


Figure 26: Sheet 1 after adding the State dimension – Columns set to Longitude (generated) and Rows to Latitude (generated). The map shows dot markers for US states where breaches occurred. State is added to the Marks card as Detail.

Adding the State dimension automatically creates a geographic map visualization. Tableau recognizes the State field as a geographic dimension and sets Columns to Longitude (generated) and Rows to Latitude (generated). The map shows scattered dots representing the US states where healthcare breaches occurred, primarily clustered in the eastern and southern United States.

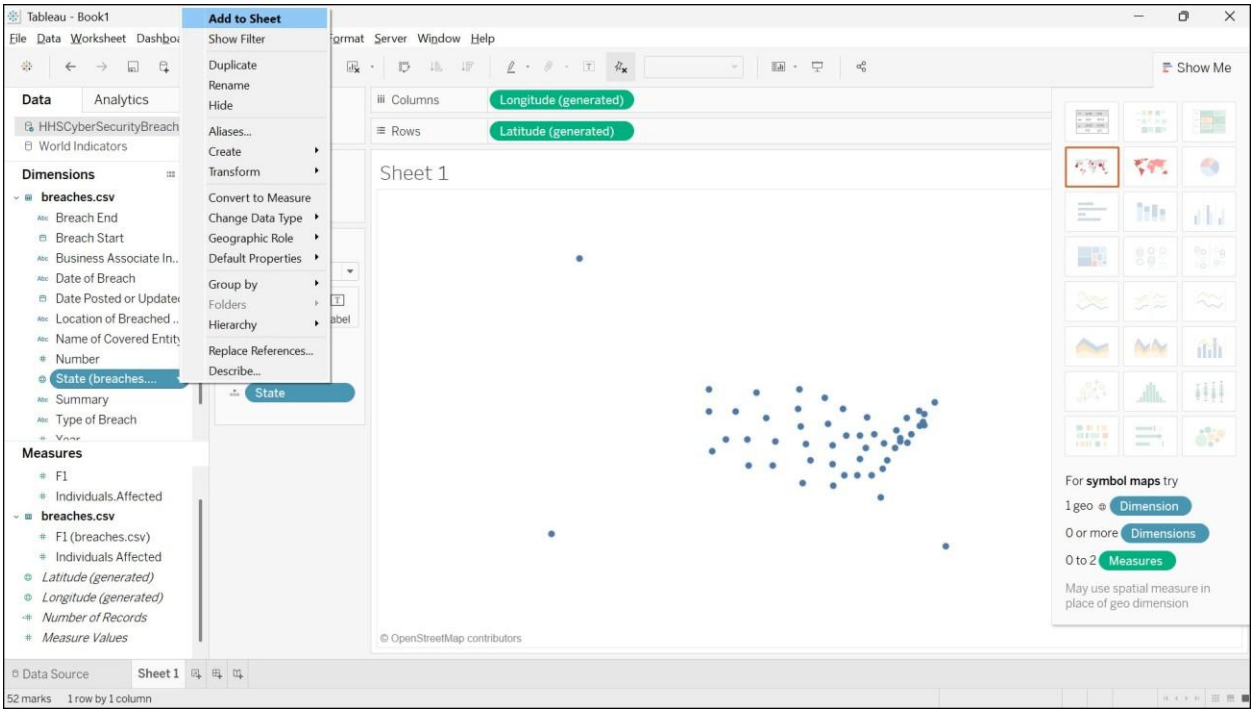


Figure 27: The State (breaches.csv) right-click context menu showing "Add to Sheet" highlighted. This adds the second State dimension from the breaches.csv dataset to the existing map visualization.

The State (breaches.csv) field from the breaches.csv table is also right-clicked and "Add to Sheet" is selected. This adds the second source's state data to the visualization, providing a combined geographic view from both datasets.

Section 1.12 – Individuals Affected Visualization (Packed Bubbles)

To visualize which states had the most individuals affected by data breaches, the Individuals Affected measure is added and the Packed Bubbles chart type is selected from the Show Me panel.

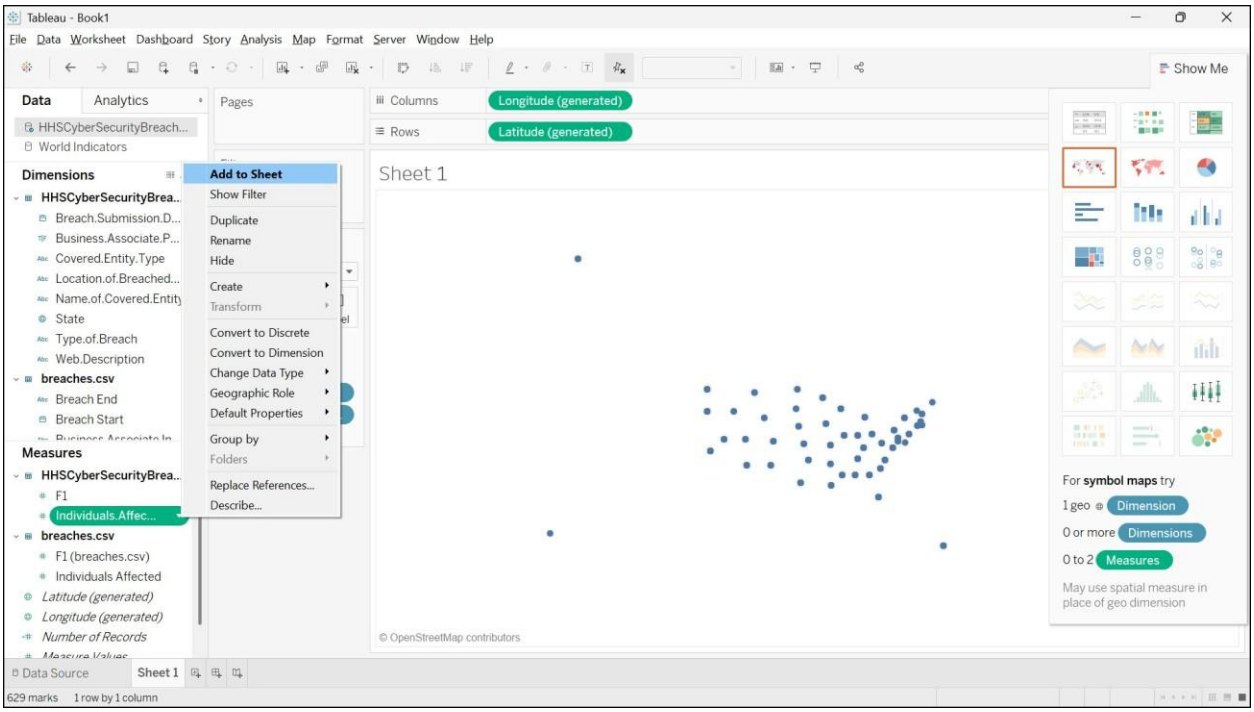


Figure 28: The Individuals.Affected measure right-click context menu showing "Add to Sheet" highlighted. The Show Me panel on the right shows the packed bubbles chart highlighted as the suggested visualization type.

The Individuals.Affected measure from HHSCyberSecurityBreaches is right-clicked and "Add to Sheet" is selected. In the Show Me panel on the right, the packed bubbles chart is highlighted as the recommended visualization for 1 or more Dimensions and 1 or 2 Measures. The chart shows 629 marks with a SUM of 37,773,276 total individuals affected.

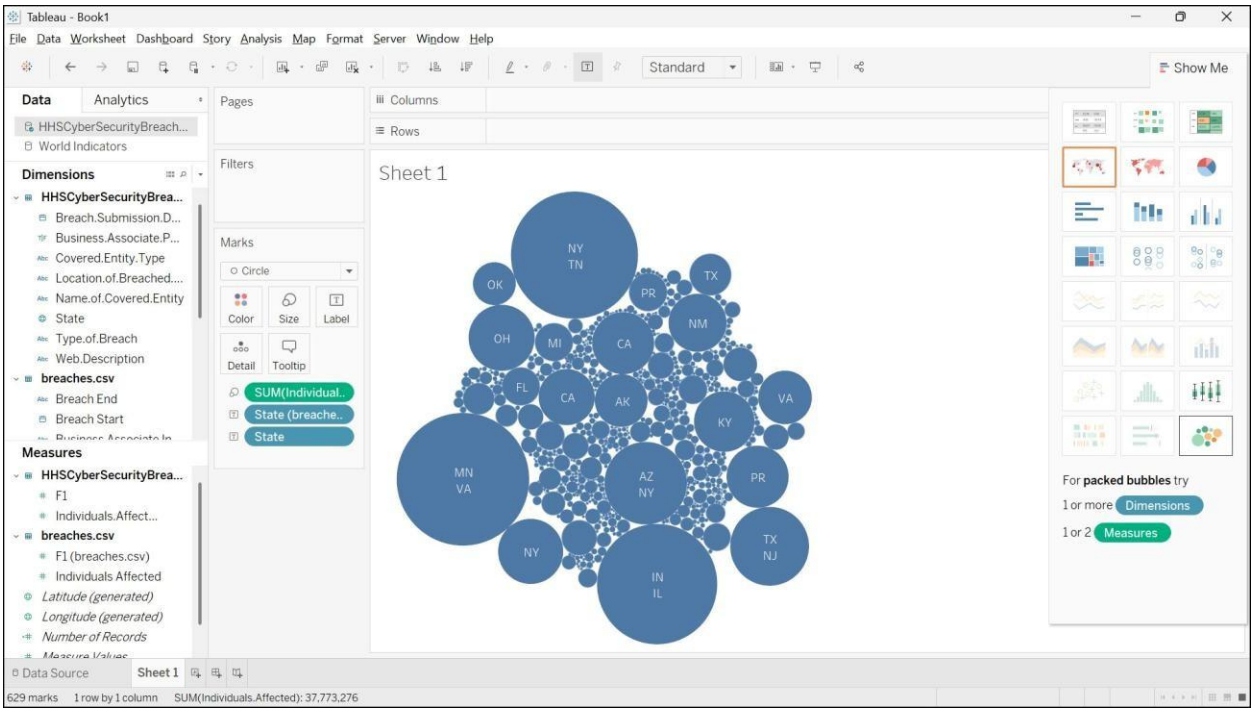


Figure 29: Packed Bubbles chart showing Individuals Affected by State. Larger bubbles represent states with more individuals affected. Notable large bubbles: NY/TN (top center), IN/IL (bottom center), TX (upper right), MN/VA (left side). Total SUM = 37,773,276 individuals affected.

The Packed Bubbles chart provides a clear visual representation of the scale of healthcare data breaches by state. The larger the bubble, the more individuals were affected. Key observations: New York (NY) and Tennessee (TN) share a prominent combined bubble near the top. Indiana (IN) and Illinois (IL) show a large bubble at the bottom, indicating significant breach impact. Texas (TX) and Virginia (VA) also show notably sized bubbles. The Marks card shows SUM(Individuals.Affected), State (breaches.csv), and State as active marks.

Section 1.13 – Stacked Bar Chart Visualization

The Stacked Bars chart type is selected from the Show Me panel to provide an alternative view of the Individuals Affected data by state.

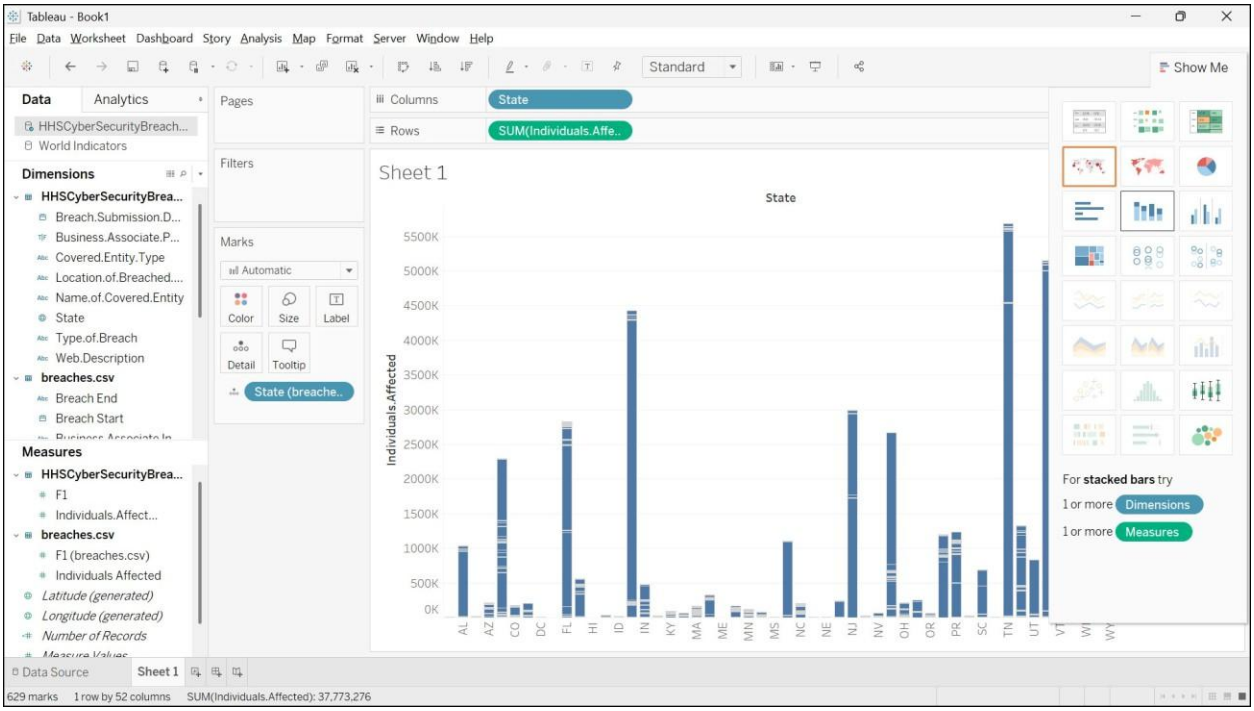


Figure 30: Stacked Bar Chart showing SUM of Individuals Affected per State. The X-axis shows state abbreviations (AL, AZ, CO, DC, FL, HI, ID, IN, KY, MA, ME, MN, MS, NC, NE, NJ, NV, OH, OR, PR, SC, TN, UT, VT, WY). The Y-axis shows values up to 5,500K. Tennessee (TN) shows the tallest bar, followed by Indiana (IN/IL group).

The Stacked Bar chart provides a detailed breakdown of individuals affected by state. Key findings from the visualization: Tennessee (TN) has the highest bar, indicating the largest total number of individuals affected by healthcare data breaches. Indiana (IN) and Illinois (IL) also show significant impact. Florida (FL) and New Jersey (NJ) show notable breach impacts. The chart confirms the findings from the packed bubbles visualization with a more precise comparative view. Total SUM(Individuals.Affected) = 37,773,276.

Part 2: Data Visualization Using Qlik Cloud Analytics

Qlik Cloud Analytics is a modern, AI-powered cloud analytics platform. In this part, the lab is repeated using Qlik to demonstrate an alternative approach to bulk data visualization. Qlik provides an associative analytics engine that uncovers hidden data relationships.

Section 2.1 – Accessing the Qlik Website and Trial Options

The Qlik website offers two main products: Qlik Talend Cloud Trial (for data integration and quality) and Qlik Cloud Analytics Trial (for analytics and visualization). The Analytics trial is selected.

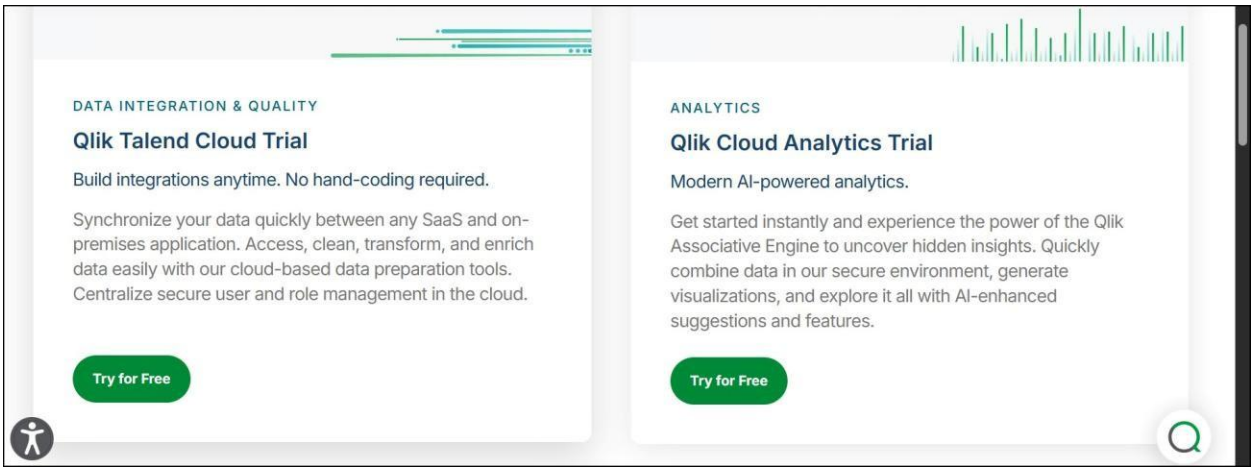


Figure 31: Qlik website product trial page showing two options: (Left) Qlik Talend Cloud Trial – "Build integrations anytime. No hand-coding required." with "Try for Free" button; (Right) Qlik Cloud Analytics Trial – "Modern AI-powered analytics" with "Try for Free" button.

The Qlik product page presents two trial options. The Qlik Cloud Analytics Trial is selected as it provides the capabilities needed for data visualization: quickly combine data in a secure environment, generate visualizations, and explore with AI-enhanced suggestions and features. The "Try for Free" button under Qlik Cloud Analytics is clicked.

Section 2.2 – Creating a Qlik Account (Password Setup)

After initiating the trial, the account setup process requires creating a password that meets the specified security requirements.

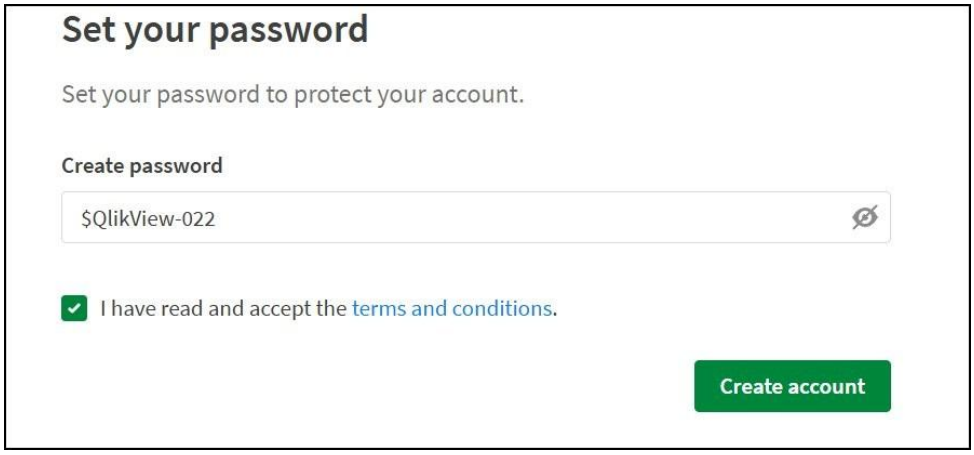


Figure 32: Qlik "Set your password" page – Password field showing "$QlikView-022", Terms and Conditions checkbox checked, and "Create account" button.

The Set your password page is displayed. A password "$QlikView-022" is entered in the Create password field. The "I have read and accept the terms and conditions" checkbox is checked. The "Create account" button is clicked to complete the account creation.

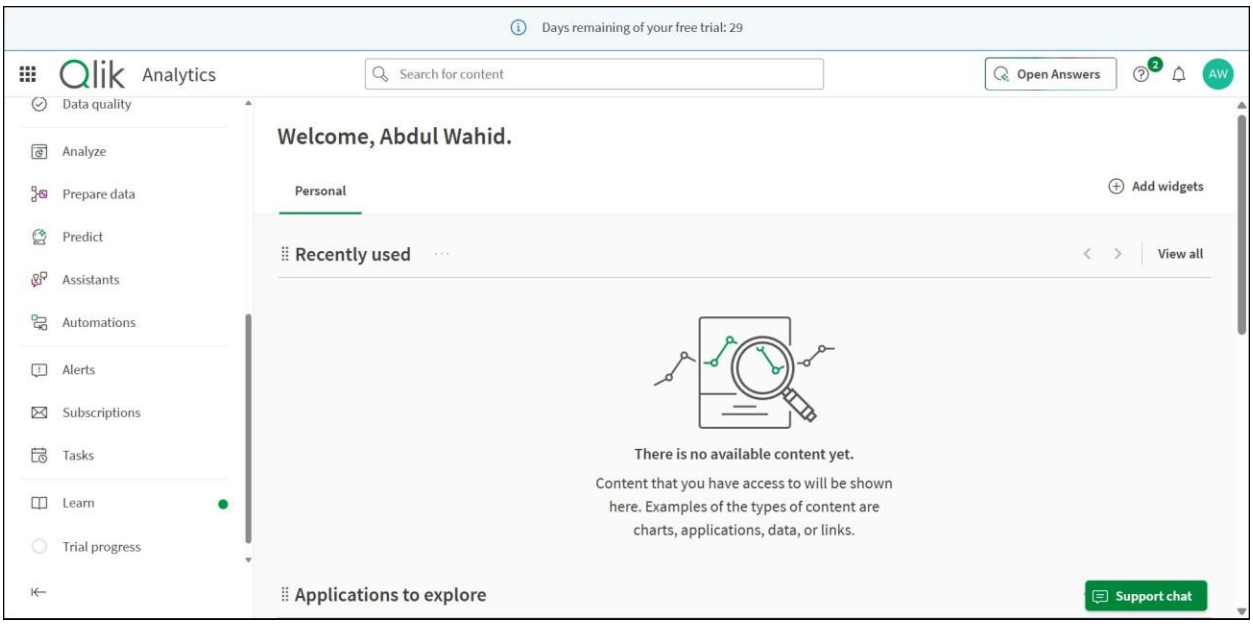


Figure 33: Qlik Cloud Analytics Trial activation page – "Almost there! Activate your trial below" with checkbox for Main Services Agreement, Product Terms Directory, and Terms of Service, and "Activate my trial" button.

After password creation, the activation page appears: "Almost there! Activate your trial below". The page shows the dashboard preview of Qlik Cloud Analytics. The user must check the agreement checkbox for Main Services Agreement, Product Terms Directory, and Terms of Service, then click "Activate my trial" to proceed.

Section 2.3 – Qlik Cloud Analytics Home Dashboard

After activation, the Qlik Cloud Analytics home dashboard opens. The user is welcomed by name and the interface shows the personal workspace.

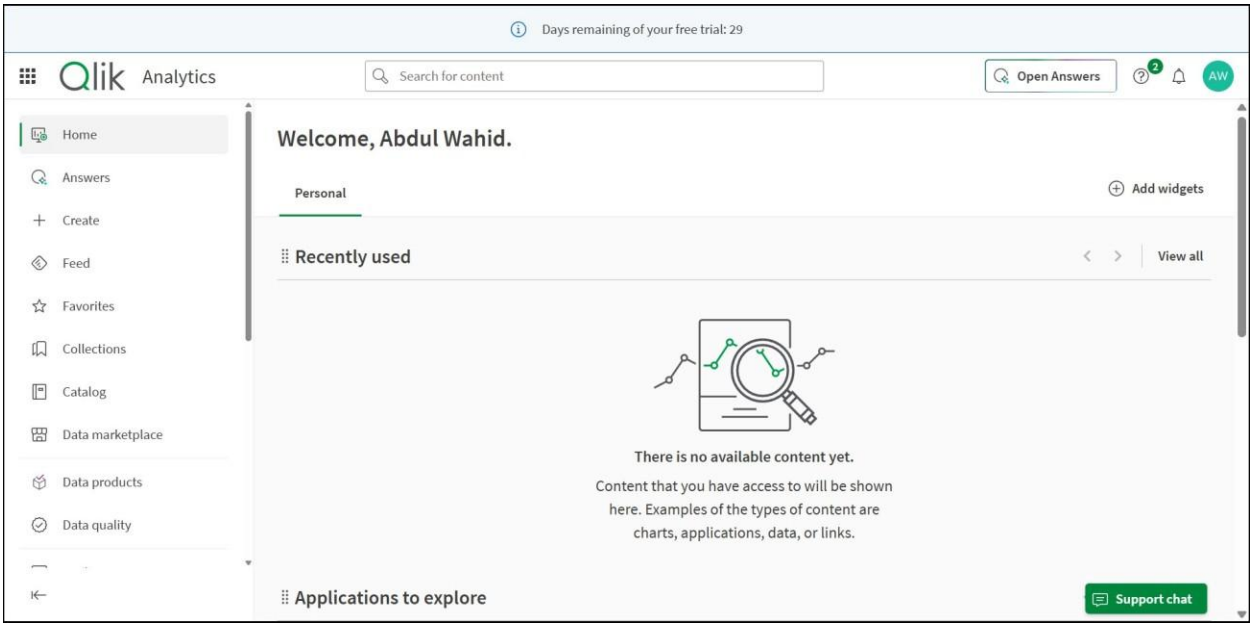


Figure 34: Qlik Cloud Analytics Home dashboard – "Welcome, Abdul Wahid." The Personal tab shows "There is no available content yet." The left navigation includes Home, Answers, Create, Feed, Favorites, Collections, Catalog, Data marketplace, Data products, and Data quality. A banner shows "Days remaining of your free trial: 29".

The Qlik home dashboard displays a personalized welcome message: "Welcome, Abdul Wahid." The interface shows 29 days remaining in the free trial. The left navigation panel provides access to key areas: Home, Answers (AI Q&A), Create (new apps), Feed, Favorites, Collections, Catalog, Data marketplace, Data products, and Data quality. The Personal tab shows no content yet as the workspace is fresh.

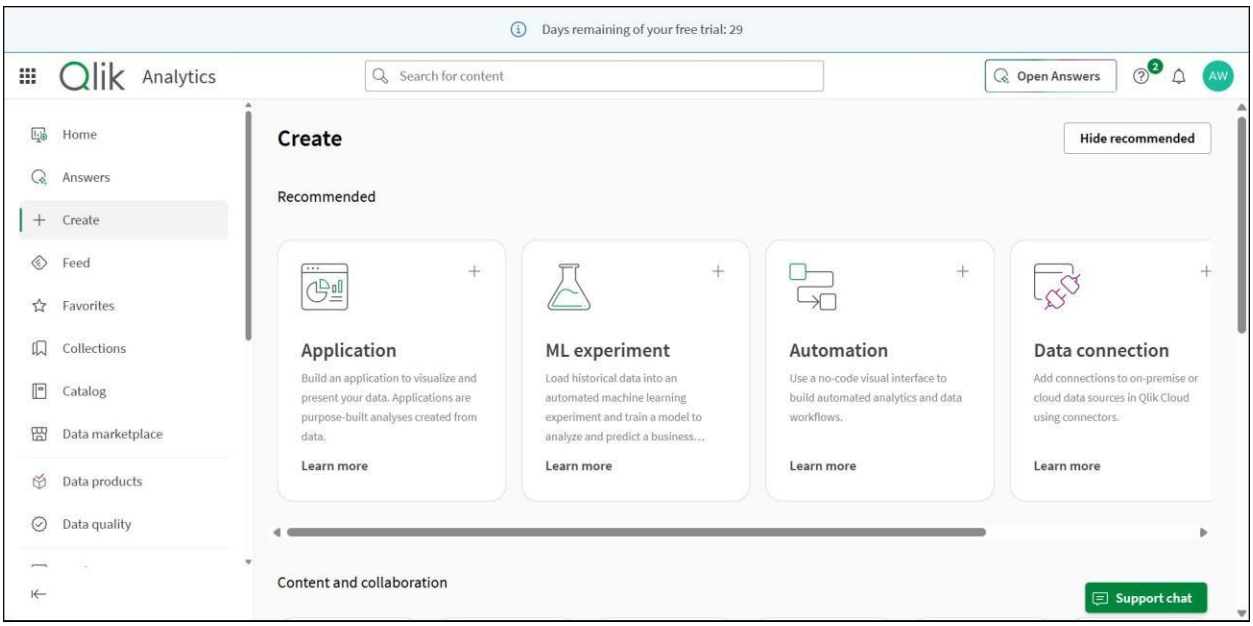


Figure 35: Qlik Create page – Shows Recommended options including Application (Build an application to visualize and present your data), ML experiment (Load historical data for machine learning), Automation (No-code visual interface for automated analytics workflows), and Data connection (Add connections to on-premise or cloud data sources).

The Create page in Qlik shows recommended creation options. For this lab, "Application" is selected – described as "Build an application to visualize and present your data. Applications are purpose-built analyses created from data." This will serve as the container for the healthcare breach analysis.

Section 2.4 – Creating a New Qlik Application

A new application named "Healthcare_Breach_Analysis" is created in the Personal space to organize the data and visualizations.

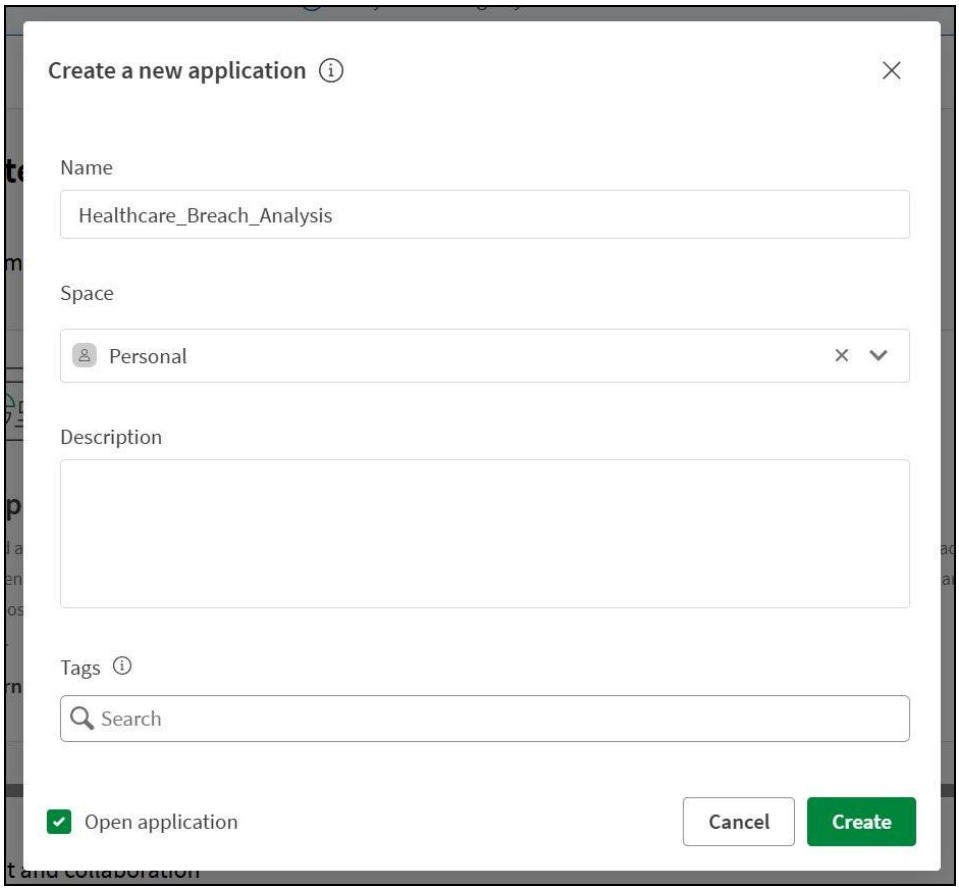


Figure 36: "Create a new application" dialog – Name field shows "Healthcare_Breach_Analysis", Space set to "Personal", Description field is empty, "Open application" checkbox is checked, and "Create" button.

The "Create a new application" dialog is filled with: Name: "Healthcare_Breach_Analysis", Space: "Personal" (selected from dropdown), Description: left empty, "Open application" checkbox: checked to immediately open after creation. The "Create" button is clicked to create and open the application.

Section 2.5 – Loading Data into Qlik

After creating the application, the data loading process begins. The HHSCyberSecurityBreaches.csv dataset is uploaded to Qlik.

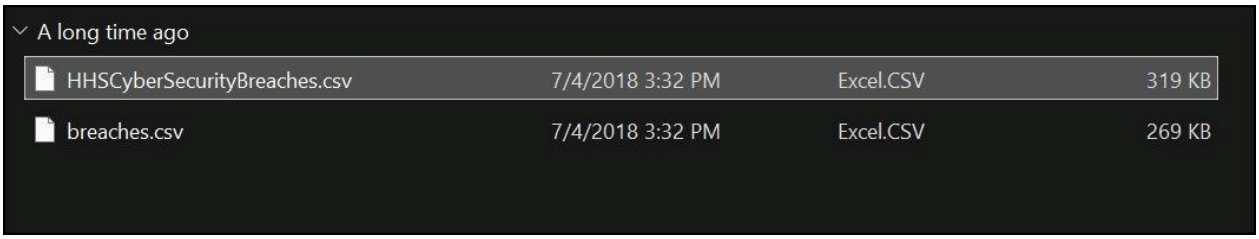


Figure 37: File selection showing the two CSV files available for upload – HHSCyberSecurityBreaches.csv (319 KB) and breaches.csv (269 KB), both dated 7/4/2018.

The same two CSV files used in the Tableau section are available: HHSCyberSecurityBreaches.csv (319 KB) and breaches.csv (269 KB). These files are selected for upload into the Qlik application.

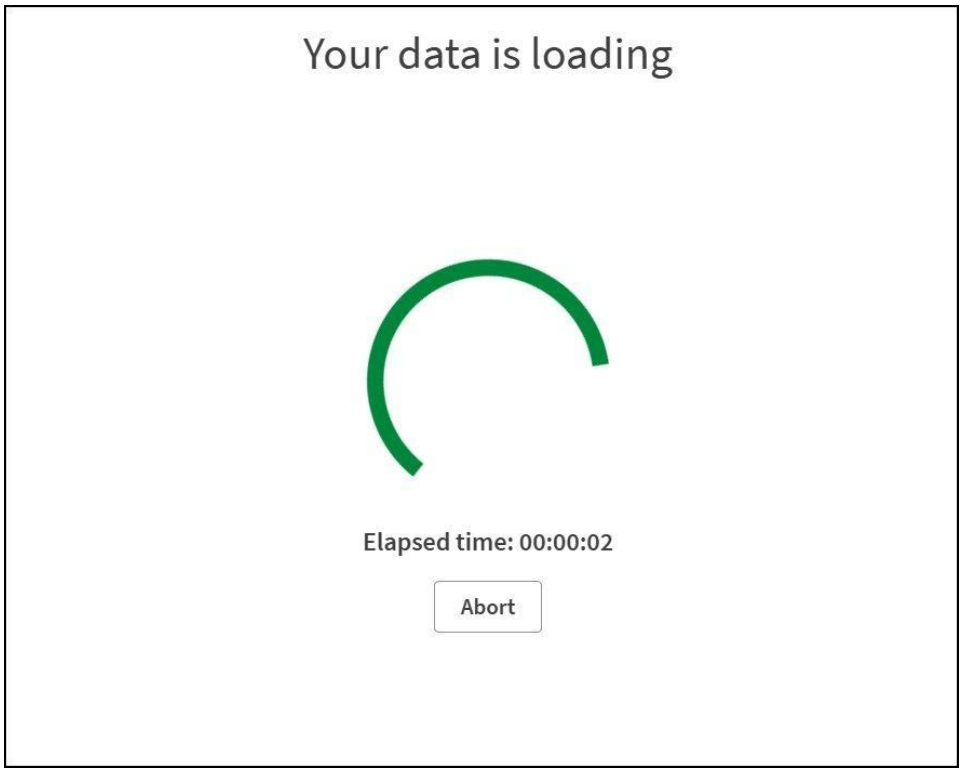


Figure 38: Qlik data loading screen showing "Your data is loading" with a circular progress indicator and elapsed time of 00:00:02. An "Abort" button is available.

The data loading screen shows a circular green progress indicator with the message "Your data is loading". The elapsed time shows 00:00:02, indicating the upload is in progress. The Abort button allows cancellation if needed. Qlik processes the CSV data and prepares it for analysis.

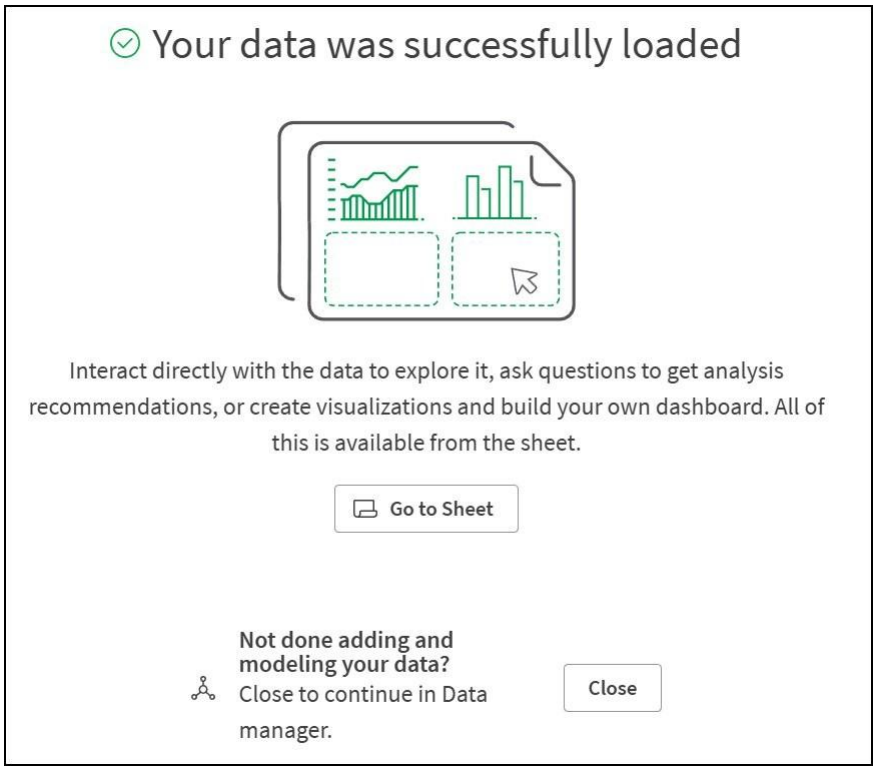


Figure 39: Qlik "Your data was successfully loaded" confirmation screen – Shows success message with options to "Go to Sheet" or "Close" to continue in Data manager.

The data loading completes successfully with the confirmation: "Your data was successfully loaded". The screen provides two options: (1) "Go to Sheet" – to start building visualizations immediately, or (2) "Close" – to continue in the Data manager for further data modeling. "Go to Sheet" is clicked to proceed.

Section 2.6 – Choosing Analytics Options in Qlik

After data loading, Qlik presents options for how to start analyzing and visualizing the data.

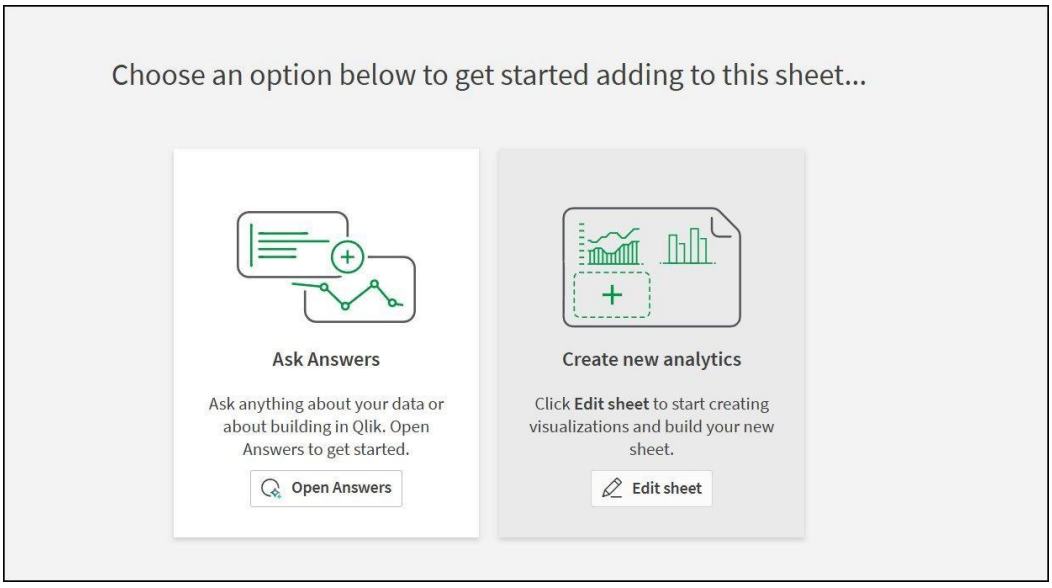


Figure 40: Qlik "Choose an option to get started" screen – Two options: (1) Ask Answers – "Ask anything about your data or about building in Qlik. Open Answers to get started." with "Open Answers" button; (2) Create new analytics – "Click Edit sheet to start creating visualizations and build your new sheet." with "Edit sheet" button.

Qlik presents two pathways: "Ask Answers" uses AI to answer natural language questions about the data, while "Create new analytics" allows manual creation of visualizations. "Create new analytics" is selected by clicking "Edit sheet" to build custom dashboards.

Section 2.7 – Qlik Sheet Editor

The Qlik sheet editor provides a canvas for creating visualizations. The left panel shows chart types, fields, and other assets.

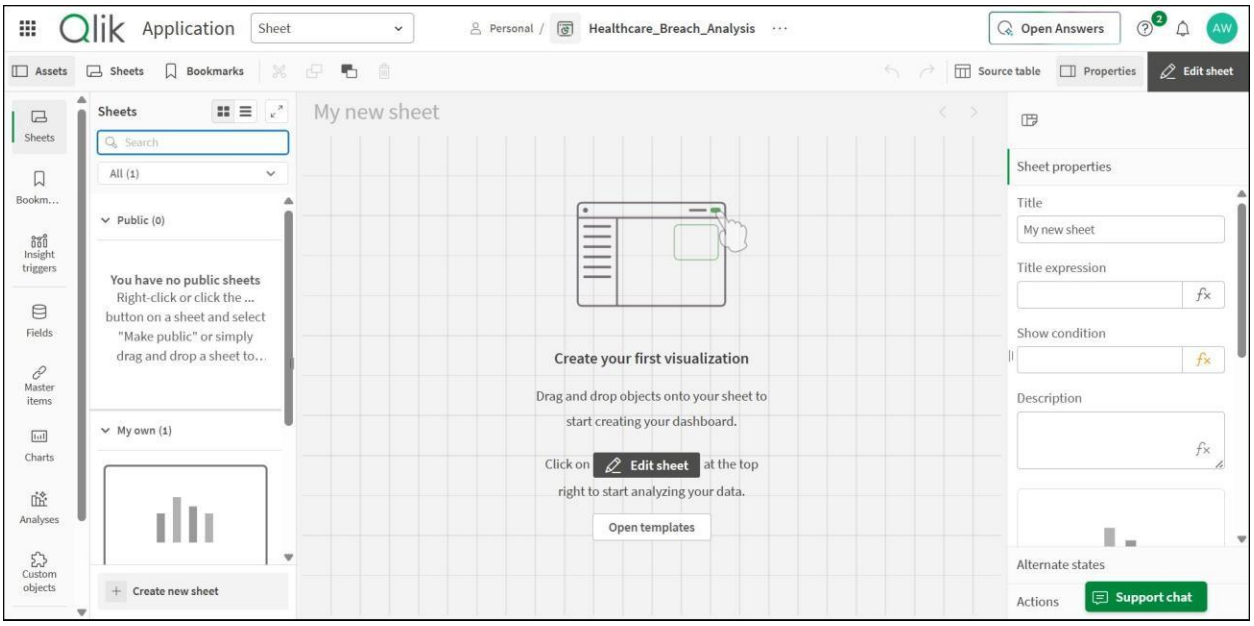


Figure 41: Qlik Healthcare_Breach_Analysis application – Sheets panel showing "My new sheet" under My own (1). The main canvas shows "Create your first visualization – Drag and drop objects onto your sheet to start creating your dashboard." Sheet properties panel is visible on the right with Title field showing "My new sheet".

The Qlik sheet editor shows the sheet canvas in the center with instructions to "Drag and drop objects onto your sheet to start creating your dashboard." The Sheets panel on the left lists the available sheet under "My own (1)". The Sheet properties panel on the right allows renaming the sheet and configuring properties. A "+ Create new sheet" option at the bottom allows adding more sheets.

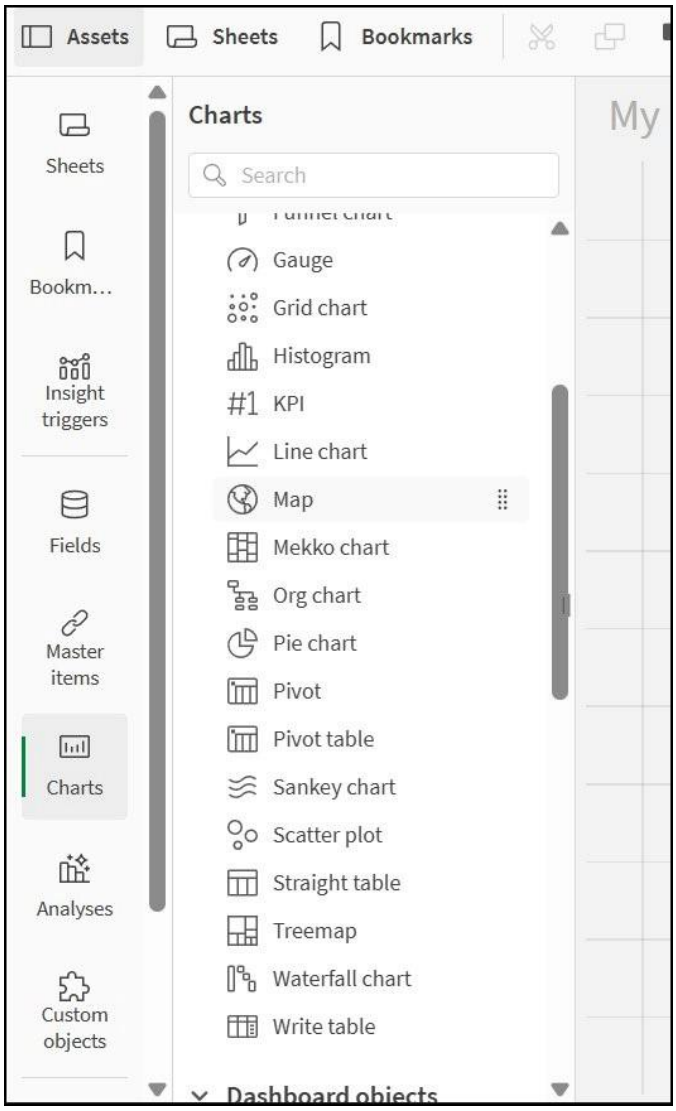


Figure 42: Qlik Charts panel showing the full list of available visualization types: Bar chart, Box plot, Bullet chart, Combo chart, Distribution plot, Filter pane, Funnel chart, Gauge, Grid chart, Histogram, KPI, Line chart, Map, Mekko chart, Org chart, Pie chart, Pivot, Pivot table, Sankey chart, Scatter plot, Straight table, Treemap, Waterfall chart, Write table, plus Dashboard objects.

The Charts panel in Qlik offers an extensive range of visualization types. For the healthcare breach analysis, the KPI chart type is used to show Total Individuals Affected, and a Map chart is used for geographic visualization. The panel also shows Dashboard objects at the bottom for more complex interactive elements.

Section 2.8 – Adding KPI Chart and Measures

A KPI (Key Performance Indicator) chart is added to the sheet to display the total count and sum of individuals affected.

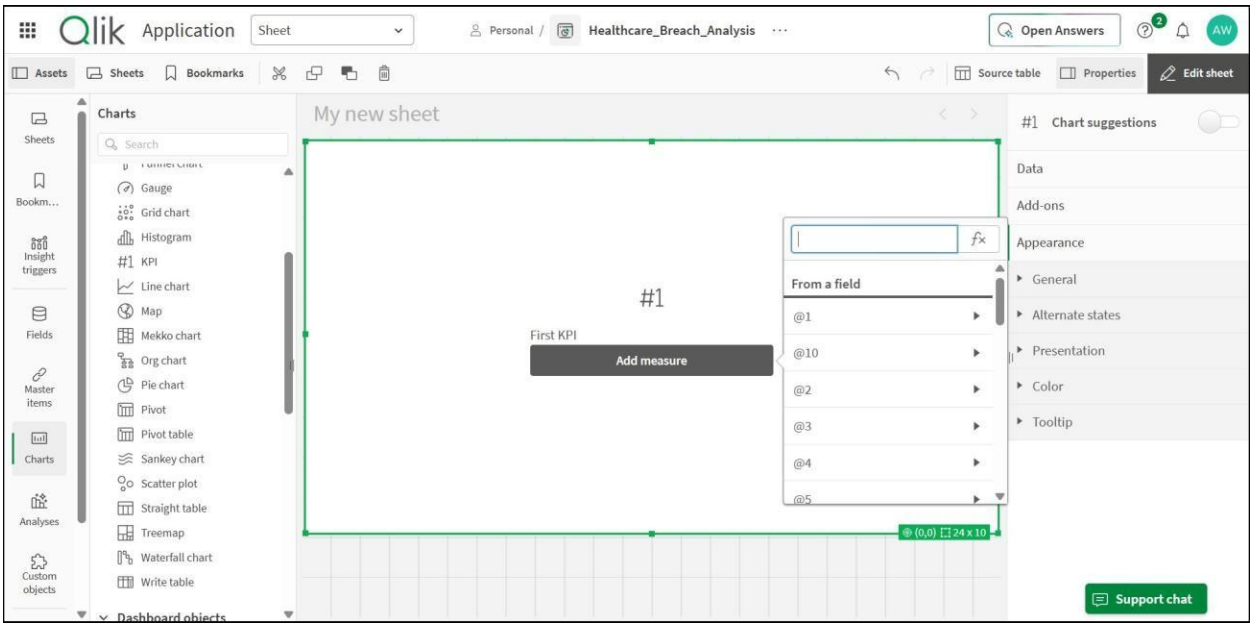


Figure 43: Qlik KPI chart editor showing the "Add measure" button active for the First KPI. The field selector dropdown shows available fields: @1, @10, @2, @3, @4, @5 from the loaded dataset. The chart currently shows "#1" as placeholder.

The KPI chart editor is opened by dragging a KPI chart onto the sheet. The field selector appears showing the available data fields from the loaded CSV (labeled @1, @10, @2, etc. as the CSV headers weren't properly named). The fields are selected to configure the First KPI (Count) and Second KPI (Sum) measures.

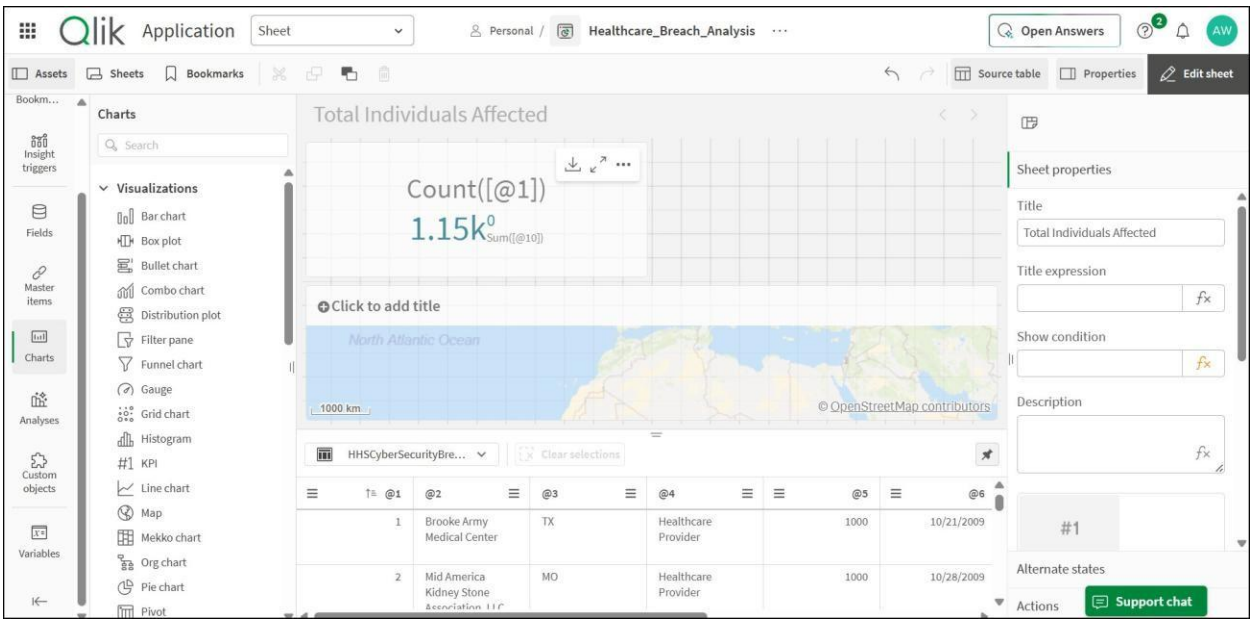


Figure 44: Qlik sheet with KPI chart showing "Count([@1]) = 1.15k" as First KPI and "Sum(@10)" as Second KPI. Below is a Map chart centered on North Africa/Atlantic, and the HHSCyberSecurityBre... data table showing the raw data records including Brooke Army Medical Center (TX, Healthcare Provider, 1000) and Mid America Kidney Stone Association (MO, Healthcare Provider, 1000).

The KPI chart now displays: Count([@1]) = 1.15k (showing approximately 1,150 breach records in the dataset) and Sum(@10) as the secondary metric. A Map chart is placed below showing the geographic

coverage. The data table at the bottom confirms the raw data is loaded correctly with the expected healthcare breach records.

Section 2.Y – Configuring Sheet Title and Properties

The sheet is renamed to "Total Individuals Affected" to clearly label the dashboard's purpose.

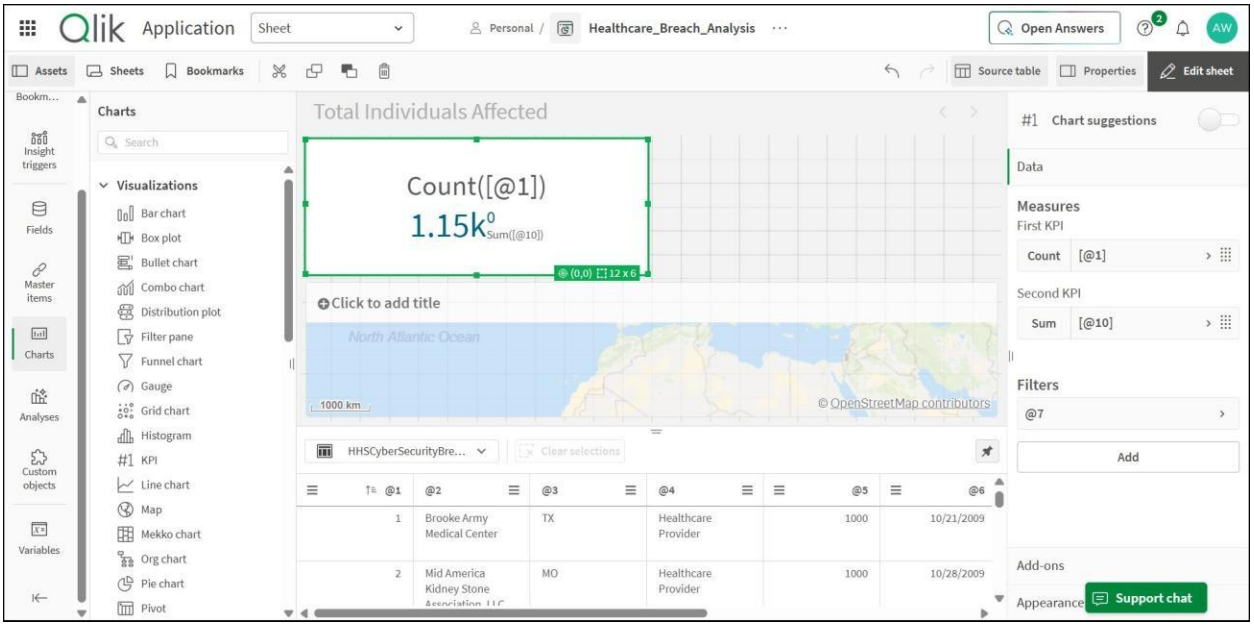


Figure 45: Qlik Sheet properties panel showing the Title field set to "Total Individuals Affected". Below the title, Title expression, Show condition, and Description fields are visible. A placeholder "#1" box is visible in the canvas area.

In the Sheet properties panel on the right, the Title field is changed to "Total Individuals Affected". This renames the sheet tab and provides a clear context for the dashboard. The Show condition and Description fields are left empty. The canvas shows a placeholder chart ready to be configured.



Figure 46: Qlik Healthcare_Breach_Analysis application header showing the sheet named "Total Individuals Affected" in the Sheet dropdown. The KPI chart shows Count([@1]) = 1.15k and the Growth trend line is visible on the right side of the KPI widget.

The final sheet view shows the "Total Individuals Affected" title in the Sheet selector dropdown at the top. The KPI widget displays Count([@1]) = 1.15k as the primary metric with a Growth trend indicator showing a relatively flat line. The application is saved in the Personal space under Healthcare_Breach_Analysis.

Section 2.10 – Qlik Cloud Analytics Capabilities

Qlik Cloud Analytics provides advanced dashboard capabilities including multi-tab sheets, trend analysis, and real-time data exploration. Below is a reference visualization showing the type of advanced analytics possible in Qlik.

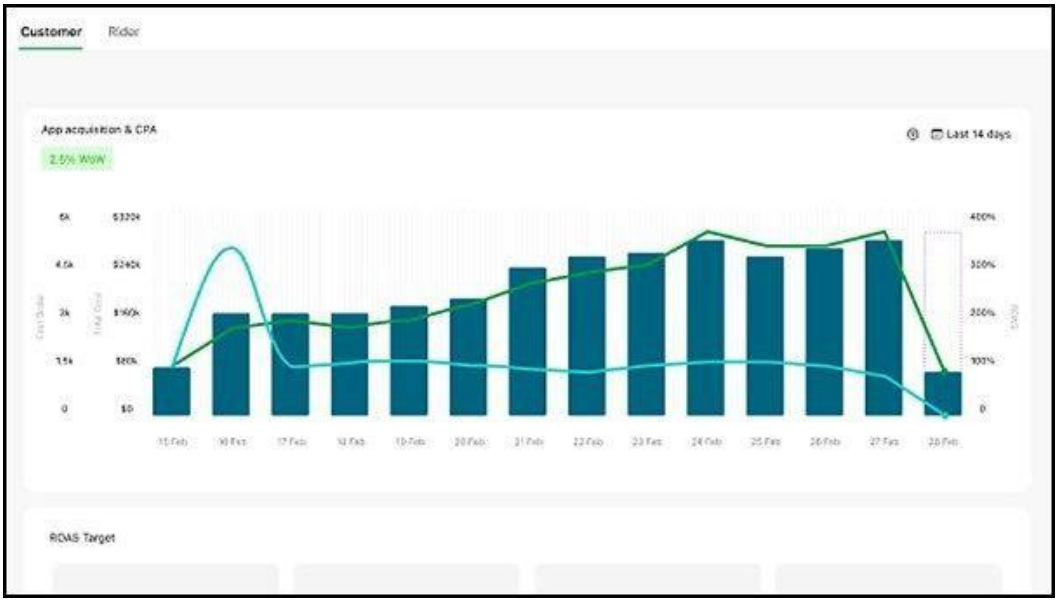


Figure 47: Sample Qlik Cloud Analytics dashboard showing "App acquisition & CPA" metrics with a combo chart (bar + line) tracking daily values from February 15-28. The chart shows two data series over a 14-day period with 2.5% WoW growth indicated. This demonstrates the advanced visualization capabilities available in Qlik.

This sample dashboard from Qlik Cloud Analytics demonstrates the platform's capabilities for trend analysis. The "App acquisition & CPA" view shows a combination chart with bar charts for daily values and trend lines overlaid. Key features visible include: 2.5% Week-over-Week (WoW) growth indicator, dual-axis chart (left: values, right: percentage), date filtering for "Last 14 days", and interactive data exploration. Such visualizations can be applied to the healthcare breach dataset for time-series analysis.

Lab Conclusion and Summary

This lab successfully demonstrated bulk data management and visualization using two industry-leading tools: Tableau Desktop and Qlik Cloud Analytics. Both tools were used to analyze the HHS Cyber Security Breaches dataset containing healthcare data breach records reported to the US Department of Health and Human Services.

Key Accomplishments

Tableau Section: Tableau Desktop 2018.1 was successfully installed on Windows 10. The trial was activated with a 14-day license after registration. Two CSV datasets (HHSCyberSecurityBreaches.csv and breaches.csv) were loaded and joined using an Inner Join on the F1 field. A data source filter was applied to show only records where Individuals Affected is between 1,500 and 2,000, resulting in 90 filtered records. Visualizations were created including a Packed Bubbles chart and a Stacked Bar chart showing individuals affected by state. Total individuals affected across all records: 37,773,276.

Qlik Section: A Qlik Cloud Analytics trial account was created with 29 days of access. The Healthcare_Breach_Analysis application was created in the Personal space. The HHSCyberSecurityBreaches.csv dataset was successfully uploaded (1,150 records loaded). A KPI chart was created showing Count = 1.15k records and a Map chart for geographic visualization. The sheet was titled "Total Individuals Affected" and configured with multiple visualization types.

Learning Outcomes

This lab demonstrated that modern data visualization tools like Tableau and Qlik provide powerful capabilities for threat intelligence analysts to: (1) Manage large bulk datasets from multiple sources, (2) Join and merge datasets for comprehensive analysis, (3) Apply filters to focus on specific data ranges, (4) Create compelling visual representations of complex data, and (5) Identify patterns and insights that would be difficult to detect in raw tabular data. The healthcare breach data analysis revealed that states like Tennessee, Indiana/Illinois, New York, and Texas had the highest impact in terms of individuals affected by healthcare data breaches.